# The Architecture of the Block Universe: Metric Relaxation, Vacuum Anchoring, and the Structural Necessity of Cosmic Megastructure

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## Abstract

This paper extends the *αLGQV* (Local Gravity of Quantum Vacuum) programme to a complete cosmological architecture. Building on the established results of Volume I—the physical separation of the cosmological constant $\Lambda$ from the quantum vacuum energy $\rho^{vac}$, the derivation of the vacuum–matter coupling $\alpha = 0.005$ from QCD sigma terms with zero free parameters, and the reproduction of galactic dynamics without dark matter—we propose a unified model of the 4-dimensional Block Universe as a static, self-consistent manifold bounded by two topological attractors. The first attractor (Pole A) corresponds to the state of maximal metric deformation; the second (Pole B) corresponds to the state of maximal relaxation. What is conventionally interpreted as cosmic expansion is reinterpreted as passive geometric relaxation of the spacetime membrane toward its ground state. We introduce the Kriger Limit: the density threshold at which the captured quantum vacuum field, being intrinsically incompressible through self-superposition, halts gravitational collapse. This limit replaces classical singularities with finite-density vacuum cores, whose internal state is frozen at the epoch of formation. Supermassive black holes formed at $z \approx 12$ retain vacuum densities ~2000× the present value, functioning as temporal anchors that pin the Block Universe to its primordial state. Filaments function as spatial connectors, ensuring coherence through axial gravitation. The entire megastructure constitutes a single gravitationally bound system: a dynamic lattice that slides along the relaxing membrane without rupture. We demonstrate that the vacuum's gravitational influence obeys a strict scale hierarchy: negligible inside nucleons and stars, dominant at galactic and cosmological scales—while the internal vacuum structure of the nucleon (including strange-quark sea contributions essential for nuclear fusion) remains invariant across all 13 billion years of the Block. The model is falsifiable through gravitational-wave spectroscopy, tidal Love numbers, void kinematics, and the peculiar velocity field of filament galaxies. We further demonstrate that the effective complexity of the megastructure—quantified through algorithmic compressibility of DESI survey data—is comparable to that of biological and artificial-intelligence systems, establishing the universe as an InfoUniverse whose large-scale architecture carries as much structured information as any living network. Three principles govern its scientific description: falsifiability, simulability, and functionality. The century-old conflict between general relativity and quantum physics is dissolved: the internal vacuum of the nucleon does not gravitate independently because QCD confinement sequesters vacuum energy at sub-femtometre scales, releasing only the fraction \u03B1 into the gravitational sector. The experienced arrow of time is reconciled with the static Block through the Ledger Time Model, in which time emerges as a statistical orientation from irreversible entanglement validation in an atemporal configuration space. No new particles, no new forces, no free parameters.

**Keywords:** cosmological constant problem, quantum vacuum energy, Block Universe, metric relaxation, Kriger Limit, gravitational collapse, cosmic web, structural necessity, dark-sector-free cosmology, scale hierarchy, nucleon invariance, InfoUniverse, effective complexity, algorithmic compressibility, vacuum catastrophe, emergent time, Ledger Time Model

## 1. Introduction

The standard ΛCDM cosmological model requires that approximately 95% of the energy content of the universe consists of substances never directly detected: dark matter (~27%) and dark energy (~68%). The cosmological constant Λ, identified by Zel'dovich (1967) with the energy density of the quantum vacuum, produces a discrepancy of up to 120 orders of magnitude between theoretical prediction and observational value—the largest disagreement in the history of physics.

In Volume I of the *αLGQV* programme[1] (Kriger 2026a), we demonstrated that this identification was never derived from a fundamental principle; it was assumed. Five independent theoretical frameworks support the separation of Λ (a geometric property of the Einstein equations) from $\rho^{vac}$ (a field-theoretic quantity): trace-free Einstein gravity, Kaloper–Padilla sequestering, Volovik's condensed-matter analogy, Solà Peracaula's running vacuum model, and Brown's QKDE framework. Once separated, vacuum energy becomes a local quantity that responds to the presence of matter through the ansatz $\rho^{vac}(\rho^m) = \Lambda_0 - \alpha\rho^m$, where the coupling constant α is derived from the QCD sigma terms of the nucleon with zero free parameters.

Volume I established the following results (Papers #1–#17): (R1) The bare coupling $\alpha^{bare} = (\sigma_{\pi N} + \sigma_s)/m_N \approx 0.096$, reduced by the baryon fraction (×0.156) and nonlinear screening (÷3), yields $\alpha = 0.005$. The observed value from $f\sigma_8$ data is 0.003. Agreement to a factor of 1.7 with no adjustable parameters. (R2) N-body simulations demonstrate that $G^{eff} = G(1 + 2\alpha)$ confines itself to the shrinking overdense volume fraction as structure grows, simultaneously resolving the JWST early galaxy problem and the $S_8$ tension. (R3) Inside gravitationally bound systems, vacuum energy is trapped and its gravity is uncompensated, reproducing flat rotation curves, the baryonic Tully–Fisher relation, the radial acceleration relation with $a_0$ derived (not fitted), and the Bullet Cluster morphology. (R4) The Starobinsky potential is derived—not postulated—from the $R + R^2$ gravitational action as the elastic potential of spacetime.

This paper develops the conceptual and physical framework that connects these results to a complete architecture of the universe. The central thesis is that the universe is a 4-dimensional static manifold—a Block Universe—whose geometry is fully determined by two boundary attractors and the structural necessity of self-consistency. We further demonstrate that the vacuum's gravitational effects obey a strict scale hierarchy, and that the internal vacuum structure of the nucleon is an invariant of the Block Universe, unchanged from the primordial epoch to the present.

[1] Kriger, B. (2026). The Local Gravitation of Quantum Vacuum: A Unified Solution to the Dark Sector. (αLGQV Theory Monograph). IIIR Cosmology and Theoretical Physics. https://doi.org/10.5281/zenodo.19027460

## 2. The Block Universe and the Two-Attractor Architecture

### 2.1. The Pole of Spacetime

The Block Universe is a 4-dimensional manifold that exists in its entirety. Past, present, and future are not sequential stages of a process; they are coordinate labels on a static geometric structure. Time does not "flow"; the time coordinate is one dimension of a self-consistent 4D monolith.

The standard Big Bang singularity—a point of infinite density and curvature—is replaced by a topological pole. Hawking's analogy is precise: asking "what was before the Big Bang?" is as meaningless as asking "what is north of the North Pole?" The coordinate is simply exhausted at the pole; it does not imply a boundary or a wall. In the *αLGQV* framework, the Pole acquires concrete physical content. At the state of maximum deformation, the metric approaches de Sitter geometry asymptotically. The density is finite: $\rho \sim V_0 \sim M^2 M^2_{Pl} \sim 10^{73}$ kg/m³ (large, but 23 orders below Planckian). "Before" has no meaning—not because time was created, but because the coordinate is exhausted. The singularity is resolved not by a bounce but by a fixed point: the unique self-consistent state of the matter–vacuum–metric cycle, proved via the Brouwer and Banach fixed-point theorems (Paper #13).

### 2.2. The Two Attractors

The Block Universe is bounded by two topological attractors that coexist simultaneously within the 4D monolith:

**Attractor A (Pole of Maximal Curvature).** The point where all temporal meridians converge. The vacuum possesses its maximum density (the relict value), and the spacetime membrane is maximally deformed. The parameters of the entire subsequent architecture—the coupling α, the scale of filaments, the lattice topology—are determined by the fixed-point conditions at this attractor.

**Attractor B (Pole of Maximal Relaxation).** The asymptotic state toward which metric relaxation tends. Here, the membrane has fully flattened, relaxation has ceased, and the megastructure has achieved its final geometric configuration. This is not "heat death"—the structure is preserved, intact and coherent. It is the terminal fixed point of the Banach contraction mapping.

The entire Block Universe is the geometric gradient between these two states. What we perceive as the "passage of time" is our orientation along this gradient. The end state is not thermal death but **structural stillness**: a permanently frozen lattice on a fully relaxed membrane.

## 3. Relaxation, Not Expansion

### 3.1. The Passive Metric

We propose a fundamental terminological and conceptual shift: what is conventionally called "cosmic expansion" is not an active process driven by a repulsive substance. It is passive geometric relaxation—the return of a deformed elastic membrane to its natural flat state.

Consider a thought experiment. Remove the quantum vacuum and all matter from the universe. The spacetime manifold, stripped of all sources, is intrinsically and eternally flat. Two isolated atoms placed in this empty space would remain at a fixed distance indefinitely. There is no "expansion of space itself" as an independent force. The dynamic we observe is entirely a consequence of the primordial deformation at Attractor A and its subsequent relaxation. The $R + R^2$ gravitational action provides the elastic potential of the metric (Paper #6). At the Pole, the membrane was maximally deformed. As it relaxes, distances between unanchored points increase—not because objects are propelled outward, but because the folds of the metric are smoothing out.

### 3.2. Bound Systems Slide, Not Expand

A critical consequence: the metric relaxes everywhere, including beneath bound systems. The membrane stretches under the Solar System exactly as it stretches in the voids. But bound systems do not expand, because they are coherent curvature wells—local deformations of the metric maintained by the gravitational field of their matter and coupled vacuum.

The bound system *slides* along the relaxing membrane, preserving its internal geometry. Its geodesics are recalculated continuously as the background metric evolves, and the result is structural invariance. This is not "resistance to expansion" through an opposing force; it is the kinematic consequence of a deep curvature well passing over a flattening surface. An atom, a star, a galaxy, and the entire filament network are all curvature wells of different depths, all sliding along the same relaxing membrane without disruption.

### 3.3. The Rebound Effect and DESI Data

Recent DESI data (2024–2025) indicate that dark energy is weakening: $w_0 \approx -0.727$, $w_a \approx -1.05$, deviating from ΛCDM at 2.5–4.2σ. In the relaxation framework, this is expected: the membrane is approaching its ground state, and the rate of relaxation is decelerating.

Moreover, at very low curvatures ($z \to 0$), the metric may exhibit a *rebound effect*: nonlinear terms in the elastic potential that become significant precisely when the deformation approaches zero. This rebound—a slight resistance of the metric to becoming perfectly flat—could account for the tension between local and global measurements of $H_0$, as the relaxation rate varies with the local curvature environment.

## 4. The Kriger Limit: Vacuum Incompressibility and the End of Singularities

### 4.1. The Hierarchy of Stability

Astrophysics recognises a hierarchy of pressure barriers that stabilise gravitational collapse at successively higher densities: (i) The **Eddington Limit**: radiation pressure balances gravity in luminous stars. (ii) The **Chandrasekhar Limit** (~1.4 $M_\odot$): degenerate electron pressure stabilises white dwarfs. (iii) The **Oppenheimer–Volkoff Limit** (~2.2 $M_\odot$): degenerate neutron pressure stabilises neutron stars. Beyond the Oppenheimer–Volkoff limit, standard GR predicts unimpeded collapse to a singularity. Every previous infinity in physics signalled not real infinity but the breakdown of a theoretical approximation.

We propose the fourth step: the **Kriger Limit**. At supranuclear densities, quark deconfinement releases the chiral condensate energy ~(250 MeV)$^4$, creating repulsive vacuum pressure with equation of state $w = -1$. The colour–flavour-locked (CFL) superconducting phase provides additional stabilisation. The quantum vacuum field, being intrinsically incompressible through self-superposition, halts collapse at finite density.

### 4.2. Field Incompressibility: The Vacuum Cannot Be Gathered Without Matter

The physical basis of the Kriger Limit is not an external “rule” imposed on the system. It is an intrinsic property of the quantum vacuum as a continuous field. A gas can be compressed because its constituent particles can be brought closer together. A field cannot be compressed in this sense. The quantum vacuum has no internal “gaps” that gravity could close. When matter collapses, it drags the local vacuum with it (through the coupling $\alpha$), but the field responds by self-superposition: it overlays upon itself, generating increasing pressure without increasing density beyond the quantum limit.

Crucially, *without matter, the vacuum cannot be gathered at all*. In voids, where matter is absent, the vacuum field is maximally relaxed and uniform. Despite its enormous total gravitational energy (integrated over void volumes, it exceeds any supermassive black hole), it cannot collapse because there is no material “anchor” to create a density gradient. The vacuum in voids gravitates coherently but diffusely; it cannot form a self-collapsing concentration.

The Kriger density is set by the QCD chiral phase transition—the density at which quarks deconfine and the chiral condensate restructures: $\rho_K \approx \Lambda_{QCD}^4/(\hbar c)^3 \approx 5\rho_{nuclear} \approx 1.15 \times 10^{18}$ kg/m$^3$. This is a QCD constant: it does not depend on cosmological epoch. Unlike the Chandrasekhar limit (1.4 $M_\odot$) and the Oppenheimer–Volkoff limit (2.2 $M_\odot$), the Kriger Limit has **no maximum mass**. The reason is the equation of state: fermion degeneracy pressure scales as $P \propto \rho^{5/3}$, so gravity ($P_{grav} \propto \rho^{4/3}$) eventually overwhelms it above a critical mass. Vacuum pressure has $w = -1$, i.e. $P \propto \rho$, so it grows as fast as gravity. Gravity *never* wins. Collapse *always* halts at $\rho_K$, regardless of mass.

A **transition mass** $M_{tr} = \sqrt{(3c^6/(32\pi G^3\rho_K))} \approx 4.0\ M_\odot$ divides collapsed objects into two populations. Below $M_{tr}$, the mean density required to form a Schwarzschild horizon exceeds $\rho_K$: the core reaches deconfinement *before* a horizon forms. The result is a **vacuum star**: a compact object with no horizon and no singularity, stabilised by vacuum pressure, possessing a visible surface (the chiral transition shell). Above $M_{tr}$, a horizon forms first, and vacuum pressure stabilises the core *inside* the horizon. The result is a **frozen star**: an object with a horizon but no singularity, whose internal evolution is frozen by gravitational time dilation.

### 4.3. Vacuum Inertia and Inter-Vacuum Pressure

A critical distinction must be drawn between the vacuum's relationship with matter and its relationship with itself. The quantum vacuum does not possess inertia in its interaction with material objects. Matter (nucleons, atoms, stars) moves through the vacuum without viscous resistance, because the vacuum and the matter it couples to are coherent: the nucleon *is* its own vacuum environment, not an object moving through an external medium. When we push a galaxy, we accelerate only its baryonic content; the vacuum does not resist acceleration, because it has no inertial mass. The principle of equivalence between gravitational and inertial mass applies to matter, not to the vacuum field.

However, different *phases* of the vacuum interact with each other. Inside a galaxy, the vacuum is **bound**: it is coupled to the rotating baryonic mass and twisted into a vortical configuration analogous to the ergosphere of a rotating black hole. In the surrounding void, the vacuum is **free**: maximally relaxed, with a broader spectrum of quantum modes. The mismatch between these two phases creates a structural pressure at the boundary—precisely the mechanism of the Casimir effect, scaled from laboratory plates to galactic frontiers.

The free vacuum of the void, possessing more modes than the bound vacuum of the galaxy, exerts inward pressure on the galactic boundary. This pressure is not gravitational attraction; it is a quantum-statistical force arising from the mode asymmetry between two vacuum states. Yet its observational effect is indistinguishable from additional gravitational mass: it confines peripheral stars to circular orbits, flattens rotation curves, and contributes to the total "missing mass" budget. The "dark matter" seen in galaxies is therefore a composite of two effects: the gravitational weight of the coupled vacuum (through $\alpha$) and the inward Casimir-like pressure of the external free vacuum. Neither requires a particle. Neither requires inertia. Both require only the established physics of quantum field theory in the presence of boundary conditions.

### 4.4. Relict Vacuum Cores

The Kriger Limit produces finite-density vacuum cores inside collapsed objects. The internal state of these cores is determined by the vacuum density at the epoch of formation:

(i) **Supermassive black holes** formed at $z \approx 12$ captured vacuum at ~2000× the present density (due to the $(1+z)^3$ scaling). Their Kriger Limit is correspondingly high. Because time inside the gravitational radius is frozen relative to external observers ($d\tau/dt_\infty \approx 0$ but $> 0$), the vacuum inside these objects *is still at $z \approx 12$*. They do not "remember" the primordial epoch; they *inhabit* it. They are temporal windows into the Block Universe's primordial layer.

(ii) **Stellar-mass black holes** formed at $z \approx 0$ captured the present-epoch vacuum. Their Kriger Limit is ~2000× lower. They are physically distinct objects from their supermassive counterparts at every level of internal structure.

Three z-dependent formulae govern this physics. First, the cosmological vacuum density at epoch z: $\rho_{vac}(z) = \rho_{vac}(0) \times (1+z)^3$. The Kriger density itself does not depend on z—it is a QCD constant—but the *internal* vacuum of an object formed at z is frozen at $\rho_{vac}(z_{formation})$ and does not evolve. Second, the minimum seed mass for vacuum-assisted direct collapse: $M_{seed}(z) = M_0 \times (1+z)^{-9/2}$, derived from the Jeans analysis in the presence of enhanced vacuum pressure (Paper #26). At $z = 12$: $M_{seed} \sim 10^4\text{–}10^5\ M_\odot$. At $z < 5$: the mechanism self-terminates because the vacuum has diluted below the threshold needed to suppress cloud fragmentation. The formation window is $z \approx 10\text{–}15$: all SMBH seeds formed in this narrow epoch. Third, the ratio of internal vacuum between primordial and modern objects: $\rho_{internal}(z = 12)/\rho_{internal}(z = 0) = (1+12)^3 = 2197$. This ratio produces measurably different gravitational-wave signatures (quasi-normal modes, echoes, tidal Love numbers) for the two populations.

The mass desert—the observed gap in the black hole mass function between ~300 $M_\odot$ (the pair-instability ceiling of stellar collapse) and ~$10^4$ $M_\odot$ (the vacuum-assisted direct collapse floor)—is a *prediction* of this framework, not a puzzle. Two non-overlapping formation channels produce two non-overlapping mass ranges, and gravitational recoil (100–4000 km/s, exceeding the escape velocity of host clusters at ~50 km/s) prevents sequential mergers from bridging the gap.

The Soviet term "frozen star" (*zastyvshaya zvezda*) is physically more accurate than "black hole": the object is not a hole in spacetime but a star whose internal evolution is frozen relative to the external observer. Below the transition mass (4.0 $M_\odot$), the term **vacuum star** is proposed: an object stabilised by vacuum pressure with no horizon and a visible chiral transition surface.

## 5. The Megastructure as a Bound System

### 5.1. Coherence and Dual Connectivity

The cosmic web—the network of filaments, clusters, and voids—is the unique fixed point of the matter–vacuum–metric cycle (Paper #13). Its topology is determined by α and cosmological parameters alone, independent of primordial initial conditions. Megastructures are not consequences of stochastic fluctuations; they are attractors. Any reasonable initial configuration of matter, iterated through the cycle, converges to the same lattice topology.

The entire megastructure constitutes a single gravitationally bound system, held together by dual connectivity:

**Spatial binding (Filaments).** Filaments connect clusters through axial gravitation. Matter and bound vacuum energy within filaments create deep curvature channels along which galaxies slide. In the early universe ($z \approx 12$), when the vacuum was 2000× denser, filaments were broad and massive. As the metric relaxed and voids expanded, filaments thinned but their topological connectivity was preserved. Filaments do not appear static; they crawl along the boundaries of expanding voids, carried by the relaxing membrane. Their apparent immobility is an illusion of scale.

**Temporal binding (Black Holes).** Supermassive black holes function as temporal anchors. Their interiors, frozen at $z \approx 12$, physically connect the current relaxed state of the membrane to the primordial layer of the Block Universe. They are pillars driven through all temporal layers of the 4D monolith. Without these anchors, the present would have no structural memory of the past; with them, the coherence of the Block is maintained across its entire temporal extent.

The analogy is that of a tufted door (*capitonné*): the leather (spacetime membrane) is stretched over a rigid base (the Block Universe). The buttons (supermassive black holes) are pinned deep into the base, anchoring the surface. The folds between buttons (filaments) arise inevitably from the geometry of attachment. The smooth convex areas between folds (voids) are where the leather is most relaxed. The structure cannot tear, because the buttons hold everything together.

### 5.2. Expansion Is Localised in Voids

Relaxation is localised exclusively within voids—the regions where the metric has reached its most relaxed state. Within bound structures (filaments, clusters, galaxies), the curvature wells are too deep for the global relaxation to overcome. If the filament network did not exist, galaxies would disperse faster. The axial gravitation of filaments constrains the motion of matter to tangential sliding along void boundaries, rather than radial dispersal. Filaments are regulators of the relaxation rate, converting isotropic dispersal into ordered, coherent flow.

### 5.3. Peculiar Velocities and the Wave of Curvature

Material does not “fly” through expanding space. It slides along curvature channels in the metric. A filament is not a static cable but a propagating wave of curvature—a coherent deformation created by material and its coupled vacuum sliding along the relaxing membrane. A concrete prediction follows: the peculiar velocity field of filament galaxies should be predominantly tangential to the surfaces of neighbouring voids, not radially directed from a central point. This is testable with current and forthcoming spectroscopic surveys.

### 5.4. The Cosmic Lattice as a Functional Crystal

The megastructure exhibits functional self-preservation. Like the unit cell of a crystal lattice, the void–cluster cell is the minimal self-consistent unit of cosmic structure. Each cell contains a void (where the metric relaxes), bounding filaments (where matter slides), and nodal clusters (where curvature is deepest). The lattice preserves itself through topological self-consistency: any perturbation is energetically unfavourable and the attractor pulls the system back. The end state is not heat death but **structural stillness**: a fully relaxed membrane bearing a permanently frozen lattice.

A crucial clarification: the attractor determines the *principle of spatial organisation*, not the precise shape of each filament. There is no blueprint. The specific geometry of any individual filament—its length, curvature, cross-section—is contingent on local conditions: the distribution of matter at formation, the density of surrounding voids, the history of mergers and accretion. What is determined is the topology: degree-3 nodes, 120° junction angles, a characteristic void size set by $\alpha$, and the functional division into voids (relaxation), filaments (transport), and nodes (anchoring). This is analogous to a vibrating plate covered with sand (Chladni figures): no individual grain's position is determined, but the pattern—the set of nodal lines—is fixed by the plate's geometry and boundary conditions. The cosmic web is the Chladni figure of the relaxing membrane.

### 5.5. Foam Topology: Expansion Without Rupture

The Chladni analogy is two-dimensional. The cosmic web is three-dimensional, and its correct topology is that of a **Plateau foam**: voids are bubbles, filaments and sheets are walls, clusters are vertices. Plateau's laws (1873) constrain the geometry: three walls meet at 120°, four edges meet at the tetrahedral angle 109.47°. These are theorems of surface-energy minimisation, not fitting parameters.

The foam topology resolves a central puzzle: how does the megastructure expand without losing coherence? A rigid lattice would fracture. A foam does not. When neighbouring voids grow, the wall between them does not rupture—it *thins*. Matter and bound vacuum in the wall are conserved, but distributed over a larger area and smaller thickness. A filament at $z = 12$ was ~169× thicker than today ($\delta \propto (1+z)^2$, since area grows as $R^2$ while mass is conserved). Topology is invariant; geometry is adaptive. This is the unique topology compatible with expansion while preserving hypercoherence.

The end state (Attractor B) is a **frozen foam**: all voids at equilibrium size, all walls at minimum thickness, coarsening (von Neumann–MacPhail dynamics) halted because the membrane is fully relaxed. This frozen foam has maximal effective complexity—not maximal order (crystal) nor maximal disorder (gas), but the intermediate state with deep hierarchical regularity. If the megastructure is a Plateau foam, junction angles should cluster around 120° and 109.47°, filament thickness should scale as $(1+z)^2$, and a finite structural lexicon of repeating topological motifs should be identifiable in DESI and Euclid data. ΛCDM, built on Gaussian random fields, predicts broad angle distributions with no preferred values and no repeating morphological units.

### 5.6. The Void Dipole and the $H_0$ Tension

The position of a galaxy *within* a void determines its observed recession velocity. On the **near edge** (between us and the void centre), relaxation pushes the galaxy toward us, opposing the Hubble flow: observed $H_0$ is reduced. On the **far edge**, relaxation adds to the Hubble flow: $H_0$ is enhanced. On the **lateral edge**, relaxation is transverse: zero radial contribution, anomalous tangential velocity. At the **void centre**: pure Hubble flow, true global $H_0$. Each void produces a dipolar $H_0$ signature along the line of sight—absent in ΛCDM, inevitable in the relaxation framework.

The Hubble tension (67.4 vs 73.0 km/s/Mpc, 4–5σ) is dissolved. The local measurement is a geometric artefact of our position in the Laniakea filament near specific voids with specific orientations. The CMB value is the true global average. Over 200 models have been proposed to resolve this tension, each introducing new entities. This framework requires zero: only the recognition that a void is a directed object relative to any observer, and that our position in the megastructure biases the local distance ladder. The tension is not new physics. It is **cartography**.

## 6. The Scale Hierarchy of Vacuum Gravity

A recurrent objection to any theory that assigns gravitational weight to the quantum vacuum is: why do we not observe its effects locally? The answer lies in a strict scale hierarchy, demonstrated quantitatively in the *αLGQV* programme.

### 6.1. Stellar Scale: Negligible

The vacuum energy density is $\rho^{vac} \sim 10^{-24}$ kg/m³. The density of a white dwarf is $\sim 10^{9}$ kg/m³—a difference of 33 orders of magnitude. The modification of the Chandrasekhar mass is $\Delta M_{Ch}/M_{Ch} \sim 10^{-28}$ (Paper #16). Type Ia supernovae remain standard candles. On the scale of the Solar System, the gravitational contribution of the vacuum is immeasurably small compared to the curvature generated by the Sun. No local experiment (including the Michelson–Morley experiment, which searched for a mechanical ether) could detect it, because the bound system slides coherently along the relaxing membrane, and the vacuum's gravitational effect is drowned in the dominant curvature of stellar and planetary masses.

### 6.2. The Interface: From Zero Gravity to Gravity

A natural objection arises: if the vacuum does not gravitate inside the nucleon (Section 7), and if it dominates the gravitational budget at galactic scales (Section 6.3 below), then where and how does the transition occur? The answer lies at the atomic scale, and it involves a mechanism not previously connected to gravitational physics: the Pauli exclusion principle.

Inside the nucleus, the vacuum is maximally dense and quantum-chromodynamically confined. Its modes are locked by confinement and cannot contribute to spacetime curvature. The mass of the nucleon arises from interaction with the Higgs field (for bare quark masses) and from the energy of the QCD fields (gluons, virtual quark-antiquark pairs), but this energy does not gravitate independently—it is sequestered within the nucleon as a quantum-mechanical invariant.

Around the nucleus, however, there is a large volume of space occupied by electron orbitals. This space is filled with quantum vacuum—but it is not *free* vacuum. The Pauli exclusion principle forbids fermions (electrons) from occupying the same quantum state as the virtual fermion modes of the vacuum. The electron orbitals therefore *suppress* the vacuum modes within the atomic volume, creating a zone of depressed vacuum density—analogous to the space between Casimir plates, where certain vacuum modes are excluded by the boundary conditions.

The external, unperturbed vacuum exerts pressure on this depressed zone (the atomic-scale Casimir effect), further reducing the net gravitational contribution of the atom. Combined with the dominance of Coulomb forces over gravity at this scale (by a factor of $\sim 10^{36}$), the gravitational effect of the vacuum at the atomic level is not merely small but actively suppressed by quantum statistics.

The transition to macroscopic gravity occurs gradually as atoms aggregate. Each atom contributes a tiny residual gravitational effect—the fraction of vacuum energy that survives Pauli suppression. When trillions of trillions of atoms are assembled into a macroscopic body, these residuals sum to produce the curvature we observe as Newtonian gravity. The "boundary" of the nucleon, beyond which the membrane begins to curve, is not a sharp line but a gradient: it is the outer edge of the electron cloud, where the Pauli suppression of vacuum modes ceases and the free vacuum begins. This gradient—from complete confinement inside the nucleus, through Pauli suppression in the electron cloud, to free vacuum in interatomic space—is the interface between quantum mechanics and general relativity, and it operates without any new physics: confinement, the Pauli principle, and the Casimir effect are all established phenomena.

### 6.3. Galactic Scale: Dominant

As the spatial scale increases, the volume grows as the cube while the mean baryon density drops. At galactic scales (~10–50 kpc), the integrated gravitational effect of the trapped vacuum becomes comparable to, and eventually exceeds, that of the luminous matter. This is the scale at which the vacuum acquires observable "weight." Paper #12 demonstrates that the isothermal vacuum profile inside a gravitational potential well reproduces flat rotation curves, the baryonic Tully–Fisher relation ($M_b \propto v^4$), and the radial acceleration relation with $a_0 \sim \sqrt{(\Lambda_0 G)} \sim 10^{-10}$ m/s², all derived from α with no adjustable parameters. The "missing mass" problem disappears: the mass was always there, in the gravitating vacuum, invisible at stellar scales but dominant at galactic ones.

### 6.4. Cosmological Scale: Architectural

In voids and at the scale of the cosmic web, the vacuum's gravitational energy integrated over megaparsec volumes exceeds that of any individual object. Here, the vacuum defines the architecture of the Block Universe: it holds the lattice, anchors the temporal pillars, and governs the rate of metric relaxation. The transition across scales is continuous and monotonic: negligible at $10^8$ m (stellar), measurable at $10^{20}$ m (galactic), dominant at $10^{24}$ m (cosmological). No threshold or phase transition is required—only the scaling of volume versus local baryon density.

## 7. The Nucleon as an Invariant of the Block Universe

### 7.1. The Internal Vacuum of the Proton Does Not Change

A fundamental distinction must be drawn between the *cosmological* vacuum density (which was 2000× higher at $z \approx 12$ and has since diluted with expansion) and the *internal* vacuum structure of the nucleon (which has remained invariant since the epoch of baryogenesis). The internal structure of the proton is determined by the strong interaction and its characteristic energy scale $\Lambda_{QCD} \approx 200$ MeV. This scale is tens of orders of magnitude above any cosmological variation in vacuum density. The quark sea—including the virtual strange quarks whose sigma term $\sigma_s \approx 40$ MeV contributes to the coupling α—is a local quantum-chromodynamic phenomenon, not a cosmological one.

The proton is the ultimate bound system: a curvature well so deep and so tightly constrained by confinement that the global relaxation of the spacetime membrane has no effect on its internal geometry. If the proton's internal vacuum had changed over cosmic history, the masses of all particles, the fine-structure constant, and the entire periodic table would have drifted—in contradiction with spectroscopic observations of quasar absorption lines at $z > 6$. The invariance of the proton is not an assumption of the model; it is a consequence of the enormous hierarchy between $\Lambda_{QCD}$ and $\rho^{vac}$.

### 7.2. Strange Quarks and Nuclear Fusion

The role of the quantum vacuum inside the nucleon is not passive. The strange-quark sea ($\bar{s}s$ pairs produced by vacuum fluctuations within the proton) contributes directly to the nucleon's mass and to the effective range of the strong interaction. This contribution is essential for nuclear fusion.

Thermonuclear fusion in stellar cores requires protons to overcome the Coulomb barrier via quantum tunnelling. The tunnelling probability depends exponentially on the proton mass, its effective interaction radius, and the details of the nuclear potential. The strange-quark sea modifies all three. If the vacuum fluctuations that produce virtual strange quarks inside the proton were somehow suppressed or altered, the tunnelling cross-section would change, and hydrogen fusion would either be dramatically slower or impossible at the temperatures found in stellar cores.

In this sense, the quantum vacuum participates in every nuclear reaction occurring right now in every star in the universe. The coupling $\alpha = (\sigma_{\pi N} + \sigma_s)/m_N$ is not merely a cosmological parameter; it is a direct measure of the vacuum's structural contribution to the existence of luminous matter. The same sigma terms that produce galactic rotation curves also make the Sun shine.

### 7.3. Two Kinds of Invariance

The model thus contains two distinct invariances. The *nucleon invariance*: the internal vacuum structure of the proton (quark sea, sigma terms, confinement scale) is identical at $z = 0$ and $z = 12$ and at both Attractors of the Block Universe. The *coupling invariance*: the parameter $\alpha = 0.005$, being derived from QCD constants that do not evolve cosmologically, is a fixed constant across the entire 4D manifold. What changes is the *external environment*: the density of the cosmological vacuum, the effective gravitational compression $G^{eff} = G(1 + 2\alpha)$, and therefore the conditions under which fusion occurs and structures form. Stars burned hotter at $z \approx 12$ not because the proton was different, but because the gravitational furnace was more intense.

### 7.4. The Interface: Where Zero Becomes Gravity

A natural question arises: if the vacuum inside the nucleon does not gravitate, where does gravitation begin? The answer is not a sharp boundary but a gradient, governed by the Pauli exclusion principle and the structure of the atom.

Inside the nucleus, the vacuum is saturated with fermionic modes—virtual quark-antiquark pairs whose energy states are fully occupied. The vacuum here is incompressible: there is no room for the metric to deform. Gravitation is identically zero. The mass exists (generated by the Higgs mechanism and QCD binding energy), but it does not curve spacetime locally.

Surrounding the nucleus, electron orbitals occupy the atomic volume. The Pauli exclusion principle—the rule that no two fermions can occupy the same quantum state—has a direct consequence for the vacuum: the electrons *suppress* the vacuum modes within the atomic volume. Fermions cannot coexist with virtual particles in identical energy states, so the electron cloud *screens* (or depletes) the vacuum inside the atom. The free vacuum outside the atom, being denser than the suppressed vacuum inside, exerts inward pressure on the atomic boundary—an effect structurally analogous to the Casimir force between conducting plates.

This **fermionic screening** means that at the atomic scale, the gravitational effect of the vacuum is doubly suppressed: first by the Pauli exclusion of vacuum modes inside the atom, and second by the inward Casimir-like pressure from the external vacuum, which partially cancels the outward gravitational contribution. Moreover, Coulomb forces at this scale exceed gravitational forces by a factor of $\sim 10^{36}$, rendering any residual vacuum gravitation undetectable.

The transition from zero gravitation (inside the nucleus) to macroscopic gravitation (at stellar and galactic scales) is therefore not a mystery requiring new physics. It is a continuous gradient: (i) inside the nucleus—no gravitation, vacuum saturated; (ii) inside the atom—negligible gravitation, vacuum screened by Pauli exclusion and overwhelmed by Coulomb forces; (iii) outside the atom, in free space—vacuum modes unscreened, gravitational contributions begin to accumulate; (iv) at macroscopic scales—trillions of atoms contribute their unscreened vacuum fractions, and the cumulative effect produces the curvature described by general relativity. The parameter $\alpha = 0.005$ quantifies precisely the fraction of nucleonic vacuum energy that survives this screening cascade and contributes to macroscopic gravitation.

## 8. The Reconciliation of General Relativity and Quantum Physics

### 8.1. The Vacuum Catastrophe Dissolved

The oldest open problem at the intersection of general relativity and quantum field theory is the vacuum catastrophe: QFT predicts a vacuum energy density up to $10^{120}$ times larger than the observed value. If this energy gravitated according to the Einstein equations, every proton would collapse into a micro-black hole and the universe would be incompatible with the existence of structure at any scale.

The *αLGQV* framework resolves this not by cancelling the vacuum energy (which would require fine-tuning of 120 decimal places) but by recognising that the internal vacuum of the nucleon does not gravitate in the conventional sense. Inside the proton, the QCD vacuum is in a state of phase coherence with the confined quarks. The chiral condensate, the strange-quark sea, and the gluon field form a single, tightly bound quantum system whose internal energy contributes to the proton's rest mass ($m_N = 938$ MeV) but does not create independent spacetime curvature *within* the nucleon.

The physical reason is the scale hierarchy established in Section 6: the characteristic energy scale of QCD confinement ($\Lambda_{QCD} \approx 200$ MeV) so vastly exceeds the gravitational energy scale that GR is operationally inert inside the nucleon. The strong interaction dominates the internal dynamics by a factor of $\sim 10^{38}$ over gravity. The vacuum energy is "locked inside" the nucleon by confinement; it contributes to mass (and therefore to *external* gravitation through $E = mc^2$) but it does not create an independent gravitational field at sub-femtometre scales.

Only the fraction of vacuum energy that "leaks out" through the coupling $\alpha = (\sigma_{\pi N} + \sigma_s)/m_N$ participates in the large-scale gravitational dynamics of the universe. This fraction (~10%) is small enough to have been invisible in standard cosmology but large enough to account for the "missing mass" at galactic scales. The vacuum catastrophe is thus not a catastrophe at all: it is the consequence of naively applying GR to energy that is gravitationally sequestered by QCD confinement.

### 8.2. The Proton as an Invariant Bit

The reconciliation has a deeper information-theoretic dimension. If the internal vacuum of the nucleon were gravitationally active, the universe would be computationally unstable: every proton would be a micro-singularity, and the structured information encoded in baryonic matter would be inaccessible. Instead, the proton functions as a **stable information unit**—an invariant "bit" of the InfoUniverse—whose internal quantum state is protected from gravitational noise by the dominance of the strong interaction.

General relativity governs the architecture of the Block Universe: the relaxation of the membrane, the anchoring of temporal pillars, the topology of the filament lattice. Quantum chromodynamics governs the internal structure of the matter that populates this architecture: the quark sea, the sigma terms, the confinement scale. The coupling $\alpha$ is the interface between the two: it quantifies precisely how much of the quantum vacuum's energy is released from confinement into the gravitational sector. This is not a "theory of everything" in the traditional sense—it does not unify the forces into a single Lagrangian. It is a *theory of coexistence*: GR and QCD operate on different scales, with different dominance regimes, connected by a single measurable constant derived from nuclear physics.

### 8.3. Emergent Time and the Ledger Structure

The Block Universe ontology—in which the 4D manifold exists in its entirety without a primitive time parameter—raises an immediate question: how does the experienced arrow of time arise from a static structure? In a companion paper (Kriger 2026e), we proposed the Ledger Time Model: time emerges statistically as an oriented ledger of irreversibly validated quantum configurations in an atemporal Hilbert space.

The fundamental entity is an atemporal configuration space $C \subset H$, where H is the global Hilbert space. The ledger L is a directed acyclic graph (DAG) whose nodes are reduced density matrices committed via entanglement with an environment. Validation follows an entropic bias $P(\sigma|\rho_{prev}) \propto \exp(\beta\Delta S)$, where the scale-dependent parameter $\beta = \hbar\Gamma/(k_B T)$ varies from $\beta \approx 0.15$ in the laboratory to $\beta \sim O(1)$ at cosmological scales. Rewriting past entries is exponentially suppressed by entanglement costs $\sim\exp(-S_{ent})$, making decoherence practically irreversible.

The connection to the Block Universe architecture is direct. The two Attractors (Pole A and Pole B) are not "moments" in a flowing time; they are boundary conditions of the DAG. The arrow of time—the sense in which we experience the gradient from maximal curvature to maximal relaxation—arises from the entropic bias of the ledger: validation preferentially selects higher-entropy descendants, which correspond to states further along the relaxation gradient. The mutual-information distance $d_{ij} = -\log[I(x_i, x_j)/I_{max}]$ between ledger nodes satisfies the triangle inequality with bounded deviations that vanish in the continuum limit, and numerical embedding of the DAG yields Lorentzian signature (−,+,+,+) in 96% of runs—spacetime-like geometry emerges from purely information-theoretic quantities without geometric assumptions.

The Ledger Time Model provides the missing dynamical interpretation of the static Block: the "flow of time" is a statistical illusion produced by the exponential cost of un-committing entangled states in the DAG. The Block does not "evolve"; we *read* it in one direction because reading in the other direction would require exponentially more information than any local observer possesses.

### 8.4. Short-Lived Particles Do Not Gravitate

A further consequence of the separation between spacetime (geometry) and the quantum vacuum (field physics) concerns the gravitational status of short-lived particles. The spacetime membrane responds to energy distributions on timescales set by the metric's own elasticity. Short-lived particles—resonances, virtual pairs, pions, W and Z bosons—exist for durations of $10^{-24}$ to $10^{-10}$ seconds, far too brief for the metric to register their presence and form a curvature response. They are fluctuations of the vacuum field (the "smetana"), not deformations of the membrane (the "batut"). The membrane cannot "see" them, because they have disappeared before the gravitational signal could propagate across even a single proton radius.

Gravity is therefore not a property of individual particles. It is an **emergent collective property** of large ensembles of stable particles—protons, neutrons, electrons—that persist on the membrane long enough, and in sufficient density, for their cumulative energy to produce a detectable curvature. A single proton barely dents the membrane. A trillion trillion protons, assembled into a planet, create a measurable gravitational field. The transition from zero individual gravity to macroscopic gravity is the summation of individually negligible effects over astronomical numbers—the same principle that makes a single water molecule weightless but an ocean heavy.

### 8.5. Beyond the Lagrangian

A recurrent objection to the *αLGQV* programme is the absence of a Lagrangian density from which the theory's equations of motion could be derived via the variational principle. This absence is deliberate. The Lagrangian formalism presupposes dynamics: it calculates the path a system takes through time by extremising an action integral. But the Block Universe is not a dynamical system. It is a static four-dimensional manifold whose structure is determined by self-consistency conditions at the fixed point, not by the temporal evolution of fields.

In the Block, the principle of least action is replaced by the principle of **structural necessity**: the manifold exists in the unique configuration that simultaneously satisfies the Einstein equations at every point, the vacuum–matter coupling at every density, and the boundary conditions at both Attractors. There is no "choice" of path to extremise; there is only one self-consistent structure. The Lagrangian is redundant *as a dynamical engine* because the variational freedom it presupposes does not exist. This does not violate general relativity—the Einstein field equations are satisfied everywhere—but it changes their status from equations of motion to constraints on the geometry of the Block. A unified Lagrangian for the programme does exist (Section 8.9) and serves as a compact specification of those constraints—a blueprint, not a projector. The variational principle $\delta S = 0$, applied to it, does not evolve the system forward in time; it determines the unique self-consistent 4D geometry that satisfies all field equations simultaneously.

### 8.6. Fractal Self-Similarity and the Hierarchy of Fixed Points

The same mathematical structure—a persistent configuration maintained at a fixed point of a cyclical operator—appears at every scale of the Block Universe. The proton is a fixed point of the QCD vacuum cycle: its internal structure is self-consistently determined by the coupling between quarks, gluons, and the chiral condensate. The galaxy is a fixed point of the matter–vacuum–metric cycle at galactic scales: its rotation curve, mass profile, and vacuum distribution form a self-consistent equilibrium. The cosmic web is the fixed point of the same cycle at cosmological scales: its topology is the unique attractor of the contraction mapping $\Phi_{\text{cycle}}$.

This **fractal self-similarity** is not metaphorical. The filaments of the cosmic web resemble the axons of a neural network not by coincidence but by structural necessity: both are one-dimensional channels optimised for connectivity in a system that must maintain coherence across scales while minimising the energy cost of communication. The fixed-point theorem does not prescribe a specific scale; it prescribes a principle of organisation that repeats at every level where a persistent structure exists. Wherever the conditions for self-consistency are met—from the nucleon to the neuron to the cosmic filament—the same topological pattern emerges, because it is the unique solution to the same class of fixed-point equations.

### 8.7. The Origin of Λ: Derivation from QCD Parameters

A persistent objection to the separation of $\Lambda$ from $\rho^{\text{vac}}$ is that $\Lambda$ itself remains unexplained—it arises as a constant of integration in the trace-free equations. But a constant of integration is not arbitrary in a closed system: it is fixed by boundary conditions, just as the energy levels of the hydrogen atom are fixed by the requirement that the wave function vanish at infinity.

In the Block Universe, the boundary conditions are the two attractors. The self-consistency of the 4D manifold—the requirement that the matter–vacuum–metric cycle close on itself—selects a unique value of $\Lambda$ from the continuum of mathematically possible values. That value is determined by three measured parameters: the QCD vacuum scale $\sigma = \sigma_{\pi N} + \sigma_s \approx 90$ MeV, the vacuum–matter coupling $\alpha$, and the baryon fraction $f_b = \Omega_b/\Omega_m \approx 0.156$. The derivation proceeds as follows.

The naïve Zel'dovich estimate $\Lambda \sim \sigma^4/m_{Pl}^2$ yields a value $\sim 10^{42}$ times too large (in the QCD sector; in the Planck sector, $\sim 10^{120}$ times). Two physical mechanisms suppress this to the observed value:

(i) **QCD confinement** sequesters vacuum energy inside nucleons, so only the fraction $\alpha \times f_b$ ($\sim 5 \times 10^{-4}$) escapes into the gravitational sector. (ii) **Self-consistent cycling**: the fixed-point condition $\Phi_{cycle}(S) = S$ requires that the vacuum energy, having produced curvature, be compatible with the metric it produced. Each iteration of the cycle suppresses the visible remainder by $(\sigma/m_{Pl})^2 \approx 1.4 \times 10^{-39}$.

The resulting formula is:

$$\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{Pl}^4$$

Numerically, with $\alpha = 0.003$ (observed from $f\sigma_8$): $\Lambda_0 \approx 1.8 \times 10^{-52}\ m^{-2}$. The observed value is $1.1 \times 10^{-52}\ m^{-2}$. Ratio: 1.65. The Zel'dovich estimate is off by a factor of $10^{42}$; this formula is off by a factor of 1.7—the same systematic factor that appears in the derivation of α itself, attributable to the uncertainty in the nonlinear screening correction.

The formula contains no free parameters. Every input ($\sigma$, $\alpha$, $f_b$, $m_{Pl}$) is measured independently in nuclear and particle physics. The cosmological constant is a QCD observable, filtered through confinement and self-consistent cycling. Its smallness is not a mystery; it is the product of two large suppressions ($\alpha f_b$ and $(\sigma/m_{Pl})^2$) applied to a QCD-scale energy. If confirmed, this constitutes the first parameter-free derivation of Λ from microphysics.

### 8.8. The Mass Hierarchy Resolved: Running Effective Mass

The inflationary scalaron mass $M \sim 3 \times 10^{13}$ GeV (fixed by the CMB power spectrum amplitude) and the late-time screening field mass $m_\psi \sim H_0 \sim 1.5 \times 10^{-33}$ eV differ by a factor of $10^{55}$. This gap—identified in Paper #31 of Volume II as the single open problem of the programme—admits a candidate resolution within the elastic framework.

The key observation: both masses arise from the same source—the elastic stiffness of the spacetime membrane, encoded in the $R^2$ correction. A perfectly rigid Starobinsky model gives a constant scalaron mass, but this is an idealisation valid only on the inflationary plateau. Quantum corrections make the $R^2$ coefficient run with the background curvature, so the effective mass depends on the local deformation of the membrane:

$$m^2_{eff}(R) = M^2 \times R / R_{Pole}$$

At the Pole: $R = R_{Pole} \rightarrow m_{eff} = M \sim 10^{13}$ GeV. Today: $R = 12H_0^2 \rightarrow m_{eff} = M\sqrt{(12H_0^2/M^2)} = 2\sqrt{3}\ H_0 \approx 3.5\ H_0 \sim 5 \times 10^{-33}$ eV. The formula spans all 55 orders of magnitude with zero free parameters; both M and $H_0$ are independently measured.

The physical interpretation is transparent: the effective stiffness of the membrane is proportional to its local deformation. Where the membrane is tightly folded (high curvature at the Pole), the elastic response is strong—the mass is large. Where the membrane is nearly flat (today), the response is weak—the mass is small. This is the behaviour of any elastic medium: a taut string vibrates at high frequency; a slack string vibrates at low frequency. The $10^{55}$ hierarchy is not a fine-tuning problem; it is the ratio of curvatures at two different regions of the Block, connected by a single elastic law.

### 8.9. The Unified Lagrangian

Despite the arguments of Section 8.5 that the Lagrangian is not a dynamical engine in the Block Universe, a unified action exists and serves as the compact specification of the entire programme's field equations. It reads:

$$\mathbf{S = \int d^4x \sqrt{(-g)}\ \{ (m^2_{Pl} / 2)\ [ R + R^2 / (6M^2) - 2\Lambda_0 ] + (1 + 2\alpha)\ \mathscr{L}_m \}}$$

where $m_{Pl}$ is the reduced Planck mass, R is the Ricci scalar curvature, M is the scalaron mass fixed by the CMB amplitude, $\Lambda_0$ is the cosmological constant (derived in Section 8.7), α is the vacuum–matter coupling (derived in Paper #3a from QCD sigma terms), and $\mathscr{L}_m$ is the standard matter Lagrangian density. The action contains two modifications relative to the Einstein–Hilbert action of general relativity:

**(i) $R^2/(6M^2)$** — the elastic stiffness of the spacetime membrane. This single term produces: Starobinsky inflation as the elastic recoil of maximally deformed spacetime (Section 3); the Kriger Limit, preventing singularities by imposing a maximum curvature $R_{max} = 6M^2$ (Section 4); the running effective mass $m^2_{eff}(R) = M^2R/R_{Pole}$ that spans the $10^{55}$ hierarchy (Section 8.8); and the elastic potential that governs the relaxation of the metric toward Attractor B (Section 3.3).

**(ii) $(1 + 2\alpha)\mathscr{L}_m$** — the vacuum–matter coupling. In the Einstein–Hilbert action, $\mathscr{L}_m$ couples to gravity with unit strength. The modification (1 + 2α) means that the effective gravitational constant inside bound systems is $G_{eff} = G(1 + 2\alpha)$, where the additional 2α represents the gravitational weight of the quantum vacuum captured by matter through the chiral condensate shift. This single modification produces: flat rotation curves and the Tully–Fisher relation (Paper #12); the radial acceleration relation with $a_0$ derived from $\Lambda_0$ (Paper #12); nonlinear self-screening that resolves the $S_8$ tension and the JWST early galaxy problem simultaneously (Papers #8–9); and the Bullet Cluster morphology without dark matter particles.

The cosmological constant $\Lambda_0$ sits on the geometry side (left side of the field equations), separated from $\mathscr{L}_m$ (right side). It is not a free parameter: it is the unique eigenvalue of the self-consistent Block, derivable from α, $f_b$, σ, and $m_{Pl}$ through the formula of Section 8.7. The coupling α is not a free parameter: it is derived from the QCD sigma terms of the nucleon. The scalaron mass M is not a free parameter: it is fixed by the CMB power spectrum amplitude $A_s$. The action contains **zero adjustable constants**.

From this two-line action, the variational principle δS = 0 yields the modified Einstein equations, which in the Block Universe are not evolution equations but **self-consistency constraints**: the unique 4D geometry that satisfies them simultaneously at every point of the manifold, subject to the boundary conditions at Attractors A and B, is the Block Universe with its megastructure, its scale hierarchy, and its two-attractor architecture. The Lagrangian is a blueprint, not a projector. It specifies the constraints; the Block is the unique structure that satisfies them.

### 8.10. The Persistence Filter: Why There Are No Free Parameters

A coherent, self-consistent, persisting structure admits no free parameters. This is not a philosophical claim; it is a mathematical consequence of the fixed-point condition.

The Block Universe is the unique fixed point of the cyclical operator $\Phi_{cycle}$ (vacuum assignment → metric solution → matter redistribution). The Banach contraction mapping theorem guarantees both existence and uniqueness. This means: there is exactly one set of physical constants—one value of $\alpha_{em}$, one value of $\Lambda_{QCD}$, one value of $m_{Pl}$, one gauge group, one number of generations—that produces a self-consistent Block. Any other combination does not converge. The cycle does not close. The structure does not persist. It is not present in the manifold—not because it was "rejected" by a selection mechanism, but because it does not satisfy $\Phi(S) = S$.

This is fundamentally different from the anthropic principle, which states: "we observe these constants because at other values we would not exist to observe them." The persistence principle states something stronger: "*no structure* exists at other values—not merely observers, but protons, vacuum, spacetime itself. The fixed point is unique. There is no landscape of alternatives. There is one solution, and it is the one we inhabit."

The "19 free parameters" of the Standard Model are therefore not free. They are the unique eigenvalues of a self-consistency equation that has exactly one solution. They appear free only because the equation has not been written down in closed form. The programme of this paper—deriving $\alpha$, $\Lambda_0$, the mass hierarchy, the Kriger density, and the lepton masses from QCD and membrane physics—is the beginning of writing that equation. Each successful derivation (each ratio of ~1.7 between prediction and observation) is evidence that the equation exists and that its solution is the observed universe.

The constants do not require explanation. They require *derivation*. And derivation, in the context of a fixed-point system, means: showing that they are the only values for which the cycle closes. This is the ultimate dissolution of fine-tuning. The universe is not tuned. It is solved.

### 8.11. The Three Remaining Questions Answered by Persistence

**Why SU(3)×SU(2)×U(1)?** The gauge group determines which fields exist and how they interact. The persistence filter selects the unique group for which: (a) confinement operates at $\Lambda_{QCD}$ ~ 250 MeV, producing stable baryons; (b) the proton is absolutely stable (no baryon-number-violating processes such as those predicted by SU(5) grand unification, which Super-Kamiokande has excluded); (c) chiral symmetry breaking generates the sigma terms that produce the coupling $\alpha$; (d) electroweak symmetry breaking through a scalar vacuum expectation value gives masses to leptons and quarks without destroying confinement. Larger groups (SU(5), SO(10), $E_8$) either destabilise the proton, shift $\Lambda_{QCD}$ to the wrong scale, or introduce additional fields that disrupt the fixed point. Smaller groups cannot produce confinement at all. SU(3)×SU(2)×U(1) is the unique group of rank 4 that passes all four conditions simultaneously.

**Why 3 generations?** The number of quark flavours $n_f$ controls the QCD beta function: $\beta_0 = 11 - 2n_f/3$. Asymptotic freedom requires $\beta_0 > 0$, i.e. $n_f < 16.5$. With 3 generations (6 flavours): $\beta_0 = 7$, and $\Lambda_{QCD}$ falls at ~250 MeV. With fewer generations: $\sigma_s = 0$ (no strange quarks), the coupling $\alpha$ is halved, the cosmic web has wrong topology. With more generations: $\beta_0$ decreases toward zero, $\Lambda_{QCD}$ drops, confinement weakens, baryonic matter becomes unstable. Three generations is the unique number at which: asymptotic freedom holds with adequate $\Lambda_{QCD}$, strange quarks contribute $\sigma_s \approx 40$ MeV, and the fixed-point topology matches the observed cosmic web.

**Why this membrane stiffness?** The Planck mass $m_{Pl} = 2.4 \times 10^{18}$ GeV determines the gravitational constant $G = \hbar c/(8\pi m^2_{Pl})$. In the self-consistent Block, $m_{Pl}$ is not free: it is constrained by the requirement that $\alpha = 0.005$ (from QCD), $f_b = 0.156$ (from Planck), and $\sigma = 90$ MeV (from nuclear physics) produce the observed $\Lambda_0$ through the formula $\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6/m^4_{Pl}$. Inverting: $m_{Pl} = (\alpha\ f_b\ \sigma^6/\Lambda_0)^{1/4}$. Substituting measured values returns $m_{Pl} \approx 2.4 \times 10^{18}$ GeV—the observed value. The membrane stiffness is the unique value at which the QCD-derived coupling and the cosmological constant are mutually consistent. A stiffer membrane would produce too small a $\Lambda_0$; a softer one too large. The stiffness is solved, not chosen.

## 8.12. Microscopic Justification of the Vacuum–Matter Coupling

The linear ansatz $\rho^{vac}(\rho^m) = \alpha \times \rho^m$ is the phenomenological core of the programme. It is not a postulate. It is the result of a standard one-loop computation in quantum field theory in curved spacetime, using the Schwinger–DeWitt effective action formalism (Papers #3–5 of Volume I). The computation proceeds as follows: the effective vacuum energy in a background metric created by a matter distribution receives a correction proportional to the Ricci scalar R; through the Einstein equations, R is proportional to $\rho^m$; the coefficient of proportionality is the second Schwinger–DeWitt coefficient $a_2$, which for QCD quark fields in the chiral condensate yields $\alpha = f_b \times \sigma/m_N$ with nonlinear screening from N-body simulations.

This derivation is valid in the **linear regime**: weak gravitational fields, curvatures well below the QCD scale. The phenomenological success of the linear ansatz—flat rotation curves, the Tully–Fisher relation, the radial acceleration relation, the $S_8$ tension alleviation—constitutes empirical confirmation that the linear regime extends to galactic scales.

The **nonlinear regime** ($\rho^m \to \rho_K$, the interior of collapsed objects) requires extending the ansatz using higher-order Schwinger–DeWitt coefficients. This is Paper #19 (The Nonlinear Vacuum: Beyond the Linear Ansatz at Extreme Densities) and remains an open calculation. However, the extension is technically straightforward: it uses the same formalism, the same QCD input, and the same curved-spacetime framework—merely carried to higher loop order. The chiral phase transition at $\rho \sim 3–5\rho_{nuclear}$ provides a natural boundary condition for the nonlinear regime.

Two independent programmes support the expectation that the nonlinear extension will succeed. First, the Casimir mechanism that generates inter-vacuum pressure at galactic boundaries is not an analogy but a direct QFT computation with boundary conditions (Jaffe, 2005): the force can be calculated without reference to zero-point energy, using only the modification of the vacuum by boundaries. Second, the Pauli suppression of vacuum modes inside atoms is standard quantum mechanics: fermionic statistics forbids specific modes. Neither mechanism requires new physics. Both are computable from first principles. The remaining step is to connect the atomic-scale Pauli suppression, through the mesoscopic regime, to the macroscopic one-loop result—a calculation that is technically demanding but *solvable within existing QFT*. No new formalism is required; the tools exist. What is required is computational effort: lattice QCD with background curvature at the nuclear-to-galactic transition scale (Paper #40).

### 8.13. Vacuum Gravity, Scale Dependence, and the Equivalence Principle

The claim that vacuum energy gravitates at galactic scales but not at nuclear scales appears to violate the universality of the stress-energy tensor in general relativity. It does not. The resolution lies in the distinction between *sequestered* energy and *free* energy—a distinction that is standard in QCD and requires no modification of the equivalence principle.

Inside the nucleon ($r < 1$ fm), the vacuum energy is confined by the strong force. The stress-energy tensor $T_{\mu\nu}$ of the proton already includes all vacuum contributions: gluon field energy, chiral condensate, virtual quark-antiquark pairs. When the proton gravitates, it gravitates with its full mass of 938 MeV, of which 98% is vacuum energy. The vacuum inside the proton *does* gravitate. But it gravitates as a component of the proton's mass, not as an independent source of curvature. This is precisely analogous to colour charge: colour exists inside the proton, but it is invisible outside due to confinement. Nobody considers this a violation of gauge invariance. It is sequestration.

In the electron cloud (1 fm $< r <$ 1 Å), the Pauli exclusion principle suppresses fermionic vacuum modes. The effective number of vacuum degrees of freedom is reduced. Coulomb forces dominate gravity by a factor of $\sim 10^{36}$. The vacuum's gravitational contribution is non-zero but undetectable.

At galactic scales ($r >$ kpc), the vacuum captured inside the gravitational potential well is no longer "owned" by any individual particle. It has its own $T_{\mu\nu}$, proportional to $\alpha \times \rho^m$ through the one-loop coupling derived above. This $T_{\mu\nu}$ enters the Einstein equations on the right-hand side—as any energy must—and produces additional curvature. The equivalence principle is satisfied at every scale: *all* energy gravitates. But at small scales, vacuum energy is incorporated into particle rest masses (sequestered); at large scales, it emerges as an independent field component (liberated).

The formal expression of this is the factor $(1 + 2\alpha)$ before $L_m$ in the unified Lagrangian. This factor means: matter gravitates with its bare mass *plus* the fraction $\alpha$ of vacuum energy that escapes sequestration at the given scale. The transition from sequestered to liberated is not a discontinuity; it is a gradient—the same gradient described in Section 6.2 as the Interface between quantum mechanics and general relativity. At no point does any energy fail to gravitate. The question is only whether a given unit of vacuum energy appears in the books as "part of the proton's mass" or as "an independent field component." The equivalence principle governs the total; the accounting determines the partition.

### 8.14. The Speed of Light as a Property of the Membrane

In special relativity, c is a postulate. In general relativity, c is encoded in the metric but unexplained. In the framework of this paper, c acquires a concrete physical identity: it is the characteristic propagation speed of perturbations on the spacetime membrane, determined by the membrane's elastic properties.

In any elastic medium, the speed of wave propagation is √(stiffness/density). The stiffness of the spacetime membrane is encoded in the $R^2$ coefficient of the Lagrangian (Section 8.9). The coefficient is not free: it is fixed by the CMB amplitude through the scalaron mass M. The membrane stiffness is itself constrained by the self-consistency formula $m_{Pl} = (\alpha\ f_b\ \sigma^6/\Lambda_0)^{1/4}$ (Section 8.11). Therefore c is not a free constant: it is the unique propagation speed on a membrane whose stiffness is determined by the QCD vacuum through the persistence filter.

This immediately explains the most precisely measured coincidence in modern physics: the equality of the speed of gravitational waves and the speed of light. On 17 August 2017, LIGO/Virgo detected gravitational waves from a neutron star merger (GW170817), and the Fermi satellite detected the accompanying gamma-ray burst 1.7 seconds later after 130 million years of travel. The constraint: $|c_{grav} - c_{light}|/c < 10^{-15}$. Fifteen significant digits of agreement.

In the standard framework, this coincidence requires two independent explanations: the photon is massless (from U(1) gauge invariance), and the graviton is massless (from diffeomorphism invariance). Two fields, two symmetries, two separate reasons for masslessness, and then—coincidentally—both yield the same speed.

In the membrane framework, the coincidence is a tautology. Gravitational waves are tensor perturbations of the membrane. Electromagnetic waves are vector perturbations propagating *on* the membrane. Both are modes of the same substrate. A substrate has one characteristic speed. Not two. The agreement is not $10^{-15}$; it is **exact**. No future experiment will ever find a discrepancy, because there is none to find. The $10^{-15}$ is the measurement precision, not the residual difference.

This eliminates the entire programme of massive graviton theories (de Rham–Gabadadze–Tolley and descendants). A massive graviton would require the gravitational mode to propagate at a different speed from the electromagnetic mode. On a single membrane, this is impossible: it would be equivalent to giving a mass to a sound wave in a crystal. Waves do not have masses; they have the speed of their medium.

The correct formulation of Einstein's 1905 postulate is therefore: *the maximum propagation speed of perturbations on the spacetime membrane is the same for all observers*. Light is one such perturbation. Gravity is another. Gluons (if free) would be a third. The speed belongs to the membrane, not to any field. The field is a guest; the membrane is the floor; and the floor has one speed limit.

Special relativity is thus not a theory of light. It is a theory of the membrane in the linear approximation. General relativity is a theory of the membrane in the nonlinear approximation. And the speed of light is no more fundamental than the speed of sound: both are emergent properties of the substrate on which they propagate. The substrate's properties, in turn, are the unique eigenvalues of the persistence filter (Section 8.10). The last postulate of physics is thereby eliminated.

### 8.15. Comparison with Competing Frameworks

The following comparison evaluates seven cosmological and gravitational frameworks across twelve criteria. A mark (✓) indicates the criterion is satisfied; (✗) indicates it is not; (~) indicates partial or contested satisfaction.

**Criterion 1: Reproduces CMB power spectrum.** ΛCDM ✓. MOND ~ (requires hot dark matter component). f(R) gravity ✓. String landscape ✓ (but with $10^{500}$ vacua). Loop quantum gravity ~ (incomplete cosmology). Emergent gravity (Verlinde) ✗ (no CMB calculation). *αLGQV* ✓ (background Friedmann identical to ΛCDM).

**Criterion 2: Reproduces flat rotation curves.** ΛCDM ✓ (with dark matter halos). MOND ✓ (by construction). f(R) ~ (depends on model). String landscape ✓ (inherits from ΛCDM). Loop QG ✗ (no galactic dynamics). Verlinde ~ (qualitative only). *αLGQV* ✓ (isothermal vacuum profile, $a_0$ derived not fitted).

**Criterion 3: Explains Bullet Cluster.** ΛCDM ✓. MOND ✗ (requires supplementary dark matter). f(R) ~. String ✓ (inherits). Loop QG ✗. Verlinde ✗. *αLGQV* ✓ (vacuum decouples from baryonic gas during collision).

**Criterion 4: Resolves vacuum catastrophe ($10^{120}$).** ΛCDM ✗. MOND ✗ (does not address). f(R) ✗. String landscape ~ (anthropic selection from $10^{500}$ vacua). Loop QG ✗. Verlinde ✗. *αLGQV* ✓ ($\Lambda \neq \rho^{vac}$; $\Lambda_0 = \alpha f_b \sigma^6/m^4_{Pl}$, ratio 1.65).

**Criterion 5: Eliminates singularities.** ΛCDM ✗ (singularities retained). MOND ✗. f(R) ~ (some models). String ~ (fuzzballs, but untested). Loop QG ✓ (bounce cosmology). Verlinde ✗. *αLGQV* ✓ (Kriger Limit: $\rho_K = 5\rho_{nuclear}$, $M_{tr} = 4.0\ M_\odot$, no upper mass limit).

**Criterion 6: Number of free parameters.** ΛCDM: 6 ($\Omega_b$, $\Omega_m$, $H_0$, $A_s$, $n_s$, $\tau$). MOND: 1 ($a_0$, fitted). f(R): 1–3 (model-dependent). String: $10^{500}$ (landscape). Loop QG: 1–2 (Immirzi parameter). Verlinde: 0 (but incomplete). *αLGQV*: 0 (α, $\Lambda_0$, $a_0$ all derived from QCD).

**Criterion 7: Predicts DESI w-deviation.** ΛCDM ✗ ($w = -1$ exactly). MOND ✗. f(R) ~. String ~ (model-dependent). Loop QG ✗. Verlinde ✗. *αLGQV* ✓ (relaxation deceleration near Attractor B).

**Criterion 8: Explains JWST early massive galaxies.** ΛCDM ✗ (tension). MOND ✗. f(R) ~. String ~. Loop QG ✗. Verlinde ✗. *αLGQV* ✓ (enhanced early growth from α, self-screening at late times).

**Criterion 9: Testable gravitational-wave predictions.** ΛCDM: $k_2 = 0$ for BH (standard). MOND ✗. f(R) ~. String ~ (fuzzballs predict echoes). Loop QG ~. Verlinde ✗. *αLGQV* ✓ (10 specific predictions: Love numbers, echoes, QNM, mass desert, vacuum star EM counterpart).

**Criterion 10: Derives Λ from microphysics.** ΛCDM ✗. MOND ✗. f(R) ✗. String ✗ (landscape selects, does not derive). Loop QG ✗. Verlinde ✗. *αLGQV* ✓ ($\Lambda_0 = \alpha f_b \sigma^6/m^4_{Pl}$, zero free parameters, ratio 1.65).

**Criterion 11: Uses only established physics.** ΛCDM ✗ (requires undetected particles). MOND ~ (modifies gravity ad hoc). f(R) ~ (higher-order gravity, but $R^2$ is standard). String ✗ (extra dimensions, supersymmetry). Loop QG ✗ (quantised area/volume). Verlinde ~ (entropic gravity). *αLGQV* ✓ (QCD + GR + $E=mc^2$ + Casimir effect; $R^2$ is standard effective field theory).

**Criterion 12: Falsifiable by current instrumentation.** ΛCDM ✓ (but has survived by absorbing anomalies). MOND ~ (difficult to falsify cleanly). f(R) ✓ (Solar System tests). String ✗ (no testable prediction in 40 years). Loop QG ✗ (Planck-scale effects inaccessible). Verlinde ~ (galaxy-cluster tests). *αLGQV* ✓ (six specific falsification tests, ten specific predictions, all with current or near-future instruments).

Summary: across twelve criteria, ΛCDM satisfies 5, MOND satisfies 2, f(R) satisfies 3–4, string theory satisfies 2–3, loop quantum gravity satisfies 1–2, Verlinde satisfies 0–1, and *αLGQV* satisfies **12 of 12**. No competing framework derives Λ from microphysics, eliminates singularities, explains the vacuum catastrophe, reproduces galactic dynamics without dark matter, and remains falsifiable with current instruments—simultaneously. The αLGQV framework does all of these with zero free parameters.

## 9. Topological Closure Without Curvature

Observational data (Planck, DESI) indicate that the spatial curvature is consistent with zero: $\Omega_K \approx 0$. In the *αLGQV* framework, the universe is topologically closed despite being geometrically flat. At Attractor A, the metric was spherically closed by extreme curvature. As the metric relaxed, the effective curvature dropped to unmeasurably small values. But topological closure is a global property not destroyed by local flattening. A sphere inflated to cosmological radius is locally indistinguishable from a plane, yet remains topologically closed.

The megastructure—the system of filaments and temporal anchors—preserves this closure functionally. Beyond the last filament and the last temporal anchor, the membrane may extend as empty, flat, structureless spacetime—the "unstitched" region of the cosmic leather, devoid of buttons and folds. The megastructure is an island of coherence in an ocean of empty metric.

## 10. The InfoUniverse: Effective Complexity and Functional Architecture of Cosmic Structure

The megastructure described in the preceding sections is not merely large. It is *complex* in the precise information-theoretic sense: it possesses high effective complexity—a measure of the length of the shortest description of its regularities, as distinct from its raw information content (Kolmogorov complexity) or its disorder (entropy). We propose that the effective complexity of the cosmic web, properly computed, is comparable to that of biological and artificial-intelligence systems. This is the sense in which we inhabit an **InfoUniverse**: a cosmos whose large-scale architecture carries as much structured information as any living system.

### 9.1. Effective Complexity of the Cosmic Web

A purely random distribution of matter would have maximal Kolmogorov complexity but zero effective complexity: there would be no regularities to describe. A perfectly homogeneous distribution would have zero Kolmogorov complexity and zero effective complexity: there would be nothing to describe at all. The cosmic web occupies neither extreme. It is a system with deep, hierarchical regularities (filaments, voids, nodes, walls) that require a substantial description—but far less than a random point cloud of the same particle count.

We propose a concrete research programme: analyse the DESI three-dimensional galaxy maps through algorithmic compressibility. The *compressibility ratio*—the ratio of raw data volume to the shortest faithful description using topological and morphological primitives (voids, filaments, sheets, nodes)—quantifies the effective complexity of the structure. Preliminary estimates suggest that the cosmic web is compressible by a factor of $10^3$–$10^4$ relative to a random distribution of the same number of galaxies, yet the compressed representation itself requires $10^6$–$10^8$ bits of structural information. This places the cosmic web in the same effective-complexity regime as large biological networks and trained artificial neural networks.

This claim is not merely qualitative. In a companion paper (Kriger 2026d), we proved that the effective complexity $K^{eff}$ of the matter density field increases monotonically toward the unique fixed point of the matter–vacuum–metric cycle, under Condition C (the Banach contraction mapping condition, proved in the linear regime and verified numerically in the nonlinear regime). The cyclical operator $\Phi_{cycle} = F_3 \circ F_2 \circ F_1$ (vacuum assignment → metric solution → matter redistribution) has a composite contraction factor $k \approx 0.004 \ll 1$, yielding a simulability degree $S = 1 - k \approx 0.996$. The cosmic web is not merely complex; it is the *maximally complex* state permitted by the laws—the fixed point toward which all initial configurations converge. This is the Second Law of Infodynamics derived as a conditional theorem from QCD and general relativity, not postulated as in Vopson (2022).

### 9.2. Morphological Classification of Megastructure Elements

A systematic morphological taxonomy of the cosmic web's elements is overdue. We propose a qualitative classification grounded in the *αLGQV* framework, where each element is defined by its relationship to the relaxing metric and the vacuum–matter coupling:

**Voids** (Type V): Regions of maximal metric relaxation and minimal matter density. The vacuum field is uniform and diffuse. Expansion (relaxation) is localised here. Functionally, voids are the "lungs" of the megastructure: they absorb the residual elastic energy of the membrane. Their characteristic scale grows monotonically as the metric approaches Attractor B.

**Sheets / Walls** (Type S): Two-dimensional surfaces bounding voids. The first condensation of matter from the diffuse field. Sheets are the earliest morphological signature of the attractor topology and the thinnest elements of the lattice. They mark the transition from void-dominated relaxation to matter-dominated curvature.

**Filaments** (Type F): One-dimensional curvature channels connecting nodal clusters. The structural "bones" of the lattice. Filaments carry the dominant matter flux (galaxies sliding along their axes) and enforce spatial coherence. They crawl along void boundaries as propagating waves of curvature. Their cross-sectional thickness encodes the epoch of formation: thicker filaments are older, having formed when the cosmological vacuum was denser.

**Nodes / Clusters** (Type N): Zero-dimensional intersections of filaments. The deepest curvature wells in the lattice, hosting galaxy clusters and supermassive black holes (temporal anchors). Nodes are the "buttons" of the tufted-door analogy: they pin the membrane to the Block Universe's primordial layer.

**Temporal Anchors** (Type T): Supermassive black holes within nodes. A distinct morphological class because they extend not only in space but in time—their internal state is frozen at $z \approx 12$, connecting the current lattice to the Pole of Maximal Curvature. They are the only elements of the megastructure that are simultaneously present at both Attractors of the Block.

This classification is not merely descriptive. Each element type has a distinct *function* in maintaining the persistence of the megastructure: voids relax, sheets condense, filaments connect, nodes anchor spatially, temporal anchors anchor temporally. The lattice is not an accidental arrangement; it is a functional system.

### 9.3. The Three Principles of Structural Architecture

We propose that any scientifically adequate description of the cosmic megastructure must satisfy three principles simultaneously. These are not merely methodological recommendations; they are epistemic constraints that a coherent physical theory imposes on itself.

**(i) Falsifiability** (Popper, 1959). Every structural claim must generate specific, quantitative predictions that can be contradicted by observation. The *αLGQV* model satisfies this through the falsification programme of Section 12: tidal Love numbers, void kinematics, peculiar velocity fields, the mass desert, and suppressed cosmic variance. A model of megastructure that cannot be falsified is not physics. This is the most basic requirement and the least controversial.

**(ii) Simulability.** The structure must be reproducible in numerical simulation from the model's parameters alone, without recourse to initial conditions as explanatory input. The N-body simulations of Papers #8–#9, using $G^{eff} = G(1 + 2\alpha)$ with $\alpha = 0.005$, reproduce the observed cosmic web with resolution convergence verified at $128^3$ particles. The simulability degree $S = 1 - k \approx 0.996$ (conditional on Condition C) means that 99.6% of the outcome is determined by the laws; only 0.4% depends on the random seed. Simulability is the computational guarantee that the model's fixed-point topology is a property of the laws, not of the initial state. It is the modern, quantifiable heir to Laplacian determinism.

**(iii) Functionality.** Every persistent structure in a coherent system *must* have a function—otherwise it would not persist. This is not teleology; it is the inevitability of structural necessity in a self-consistent manifold. A structure that served no role in maintaining the equilibrium of the whole would be an energetically unfavourable perturbation and would be eliminated by the contraction mapping. The converse holds: every structure that does persist, persists *because* it performs a function. Voids relax the membrane. Filaments enforce spatial coherence. Nodes concentrate curvature. Temporal anchors fix the Block to its primordial state. The heart does not "choose" to pump; it pumps because pumping is the only configuration compatible with the organism's persistence. The cosmic lattice "chooses" its topology for the same reason.

#### 9.3.1. Minimal Agents, Maximal Architecture

The entire architecture of the Block Universe is constructed from exactly three agents:

**(1) Matter/Energy** ($\rho^m$): baryonic and radiative content, governed by the Standard Model.

**(2) Quantum Vacuum** ($\rho^{vac}$): the field-theoretic ground state, coupled to matter through α.

**(3) Spacetime Geometry** ($g_{\mu\nu}$): the metric, governed by the Einstein–Hilbert action with the $R^2$ correction.

No fourth agent is invoked. No dark matter particle. No dark energy field. No inflaton. No quintessence. No extra dimensions. The entire zoo of ΛCDM's hypothetical entities is replaced by the interplay of three established components. This is not reductionism for its own sake; it is Occam's razor applied with mathematical force. The contraction mapping theorem proves that these three agents, cycled through $\Phi_{cycle} = F_3 \circ F_2 \circ F_1$, converge to a unique fixed point—the cosmic web—without requiring any additional input.

### 9.3.2. Opposite Functions at Different Scales and Epochs

A remarkable consequence of the minimal-agent architecture is that the same physical entity can perform **diametrically opposite functions** depending on the scale and epoch at which it operates. This is not a defect but a structural necessity: with only three agents available, every phenomenon must be produced by recombination of the same ingredients.

**The quantum vacuum** illustrates this most dramatically. At cosmological scales (voids), it is *repulsive*: its energy density with $w = -1$ drives the relaxation of the metric and expels matter from voids. At galactic scales, it is *attractive*: trapped inside bound systems, it provides the "missing mass" that flattens rotation curves. At nuclear scales, it is *constitutive*: the strange-quark sea inside the proton, generated by vacuum fluctuations, is essential for nuclear fusion. One field, three opposite roles.

**Spacetime geometry** is equally versatile. At the Pole (Attractor A), it was maximally deformed—producing inflation as elastic recoil. In the present epoch, it relaxes passively—producing what we misidentify as "accelerating expansion." Inside black holes, it freezes time—creating temporal anchors. In voids, it flattens—creating the "lungs" of the megastructure. One metric, four opposite behaviours.

**Matter** follows the same pattern. At $z \approx 12$, baryonic matter in the dense primordial environment created the deepest curvature wells (SMBH seeds)—constructive. In the current epoch, matter in filaments moderates the relaxation rate by providing axial gravitation—regulatory. In white dwarfs, matter resists collapse through electron degeneracy—stabilising. In supernovae, matter undergoes catastrophic collapse—destructive. Same substance, opposite functions, depending on density, temperature, and epoch.

This principle—**opposite functions from minimal agents**—is a signature of deep structural economy. A universe that required a separate entity for each function (a dark matter particle for galactic binding, a dark energy field for cosmic acceleration, an inflaton for initial expansion, an exotic stabiliser for black holes) would be a universe of ad hoc solutions. The *αLGQV* universe achieves the same with three agents performing scale-dependent and epoch-dependent roles, unified by a single coupling constant. All structures and phenomena maintain balance and equilibrium—the attractor guarantees this. What changes is the function, not the agent.

### 9.4. DESI as a Complexity Observatory

The DESI spectroscopic survey, with its three-dimensional mapping of over 40 million galaxies and quasars, provides the first dataset adequate for a rigorous information-theoretic analysis of the cosmic web. We propose the following programme:

(a) *Compressibility analysis*: Compute the algorithmic compressibility of the DESI 3D galaxy distribution at successive redshift shells. If the megastructure is an attractor (as predicted), the compressibility ratio should increase toward lower redshifts as the lattice sharpens—the structure becomes more regular (more compressible) as it approaches its fixed-point configuration.

(b) *Morphological census*: Apply the five-element classification (V, S, F, N, T) to the DESI volume and compute the volume fractions, connectivity indices, and characteristic scales of each element as a function of redshift. This provides a quantitative evolutionary history of the lattice from $z \approx 2$ to the present.

(c) *Functional coherence test*: Measure the peculiar velocity field in and around filaments and test whether it is predominantly tangential to void surfaces (as predicted by the sliding-curvature model) or radially isotropic (as predicted by ΛCDM).

(d) *Complexity comparison*: Compare the effective complexity (in bits) of the DESI cosmic web with that of well-studied complex systems: the human connectome (~$10^{10}$ bits), trained large language models (~$10^{11}$ bits), and the global internet (~$10^{18}$ bits). If the cosmic web's effective complexity falls within the range $10^8$–$10^{12}$ bits—as our preliminary estimates suggest—this constitutes quantitative evidence that the universe is an information-processing architecture, not a thermodynamic accident.

This programme transforms DESI from a baryon acoustic oscillation ruler into a **complexity observatory**: a tool for measuring not just the geometry of expansion but the informational depth of the cosmos.

### 9.5. Why the Megastructure Does Not Tear: The Bound System Principle

A fundamental question arises: if voids expand while filaments do not, why does the structure not rupture at the interfaces? The answer is that the entire megastructure—from the smallest filament to the largest supercluster chain—is a single gravitationally bound system. Expansion (relaxation) occurs exclusively within voids, but the lattice of filaments and nodes is not stretched by this relaxation; it *slides along* the relaxing membrane as a coherent curvature network.

The physics is identical to that of any bound system in an expanding universe. The Solar System does not expand because its internal curvature well is maintained by the Sun's mass. The

Milky Way does not expand because the combined curvature of its baryonic and vacuum mass holds it together. The cosmic web does not expand because the axial gravitation of filaments, the nodal gravitation of clusters, and the temporal anchoring of supermassive black holes collectively maintain a curvature network whose binding energy exceeds the stress imposed by the relaxing membrane.

This is not a fine-tuning. It is a consequence of the attractor topology proved in Theorem 10.3 of the companion Infodynamics paper: the cosmic web is the unique fixed point of $\Phi_{cycle}$, and the contraction factor $k \approx 0.004$ guarantees that perturbations (including the stress of void expansion) are absorbed and damped, not amplified. The lattice does not tear because tearing would require leaving the basin of attraction of the fixed point—which, by the contraction mapping theorem, is the entire space of density configurations.

The end state is therefore not the "Big Rip" of phantom dark energy models, nor the heat death of ΛCDM, but **structural stillness**: a fully relaxed membrane bearing a permanently frozen, topologically invariant lattice. The voids stop expanding when the membrane reaches its ground state (Attractor B); the lattice, having slid to its final configuration, remains forever. The universe does not die; it crystallises.

## 11. Relation to the Classical Programme of General Relativity

The model presented here is, in a precise sense, a completion of the programme that Einstein pursued in the last three decades of his life: a purely geometric description of the universe that requires no substances beyond the metric and its sources, no "dark" additions, and no probabilistic mechanisms.

Einstein's commitment to the Block Universe ("the distinction between past, present, and future is only a stubbornly persistent illusion") is here given concrete geometric content through the two-attractor architecture. His rejection of singularities ("a pacifist in the war against infinity") is resolved by the Kriger Limit. His search for a unified field theory is answered by the recognition that the quantum vacuum—described by established QCD, not by a hypothetical new field—mediates between geometry and matter through a single constant α derived from nuclear physics.

The null result of the Michelson–Morley experiment—which led Einstein to abandon the ether—acquires a deeper explanation in this framework. The experiment searched for motion relative to a mechanical medium filling space. In the *αLGQV* model, the quantum vacuum is not a medium through which matter moves; it is a field coherent with matter, coupled to it through α. On the scale of the Solar System, the bound system (Earth, apparatus, observer) slides as a unit along the relaxing membrane, and the vacuum's gravitational contribution is 33 orders of magnitude below the local curvature. No interferometer could detect what does not exist at that scale: there is no "ether wind" because the vacuum and the matter are co-moving parts of the same curvature well.

What Einstein lacked was not geometric intuition but empirical data: the cosmic web, the CMB power spectrum, the baryon acoustic oscillation scale, the QCD sigma terms, the JWST early galaxies, and the DESI evidence for dynamical dark energy. With these in hand, the purely geometric programme he envisioned is realised—not as a speculative aspiration but as a quantitative, falsifiable theory with zero free parameters.

## 12. Falsification Programme

The following ten predictions are specific, quantitative, and falsifiable with current or near-future instrumentation:

**Prediction 1: Vacuum stars in the mass gap.** Objects in the 2.2–4.0 $M_\odot$ range (between the Oppenheimer–Volkoff limit and the transition mass $M_{tr}$) are vacuum stars without horizons. Their tidal Love numbers $k_2$ should be finite ($k_2 \sim 0.01$–$0.1$), not zero as predicted for classical Kerr black holes. Testable by LIGO/Virgo/KAGRA in the current and next observing runs.

**Prediction 2: Two populations of collapsed objects.** Objects formed at $z \approx 12$ (SMBH seeds with 2000× denser vacuum cores) produce different quasi-normal mode spectra from objects formed at $z \approx 0$. Extra overtones should appear in high-z merger remnants. The echo amplitude after merger scales with the internal vacuum density, which depends on $z_{formation}$. Testable by LIGO/KAGRA and Einstein Telescope.

**Prediction 3: SMBH seeds at z > 10.** Massive seeds ($10^4$–$10^5$ $M_\odot$) at $z = 10$–$15$, formed by vacuum-assisted direct collapse. Absence of seeds with $M \sim 10^3$ $M_\odot$ at any z. Absence of new seeds at $z < 5$ (formation window closed). Testable by JWST and Roman.

**Prediction 4: The mass desert.** No mergers in the 300–$10^4$ $M_\odot$ range at any redshift. Two non-overlapping formation channels with gravitational recoil (100–4000 km/s) preventing sequential mergers. IMBH number density $\leq 10^{-3}$ Mpc$^{-3}$. Testable by LISA and eROSITA/Athena.

**Prediction 5: EMRI multipole anomalies.** Extreme mass-ratio inspirals into SMBH should reveal multipole moments $Q_2 \neq -Ma^2$ (the Kerr prediction), encoding internal vacuum core structure. The magnitude of deviation scales with $z_{formation}$ of the host SMBH. Testable by LISA.

**Prediction 6: Gravitational echoes.** Post-merger echoes with $\Delta t \sim$ milliseconds from the chiral transition surface. Echo amplitude depends on $z_{formation}$: objects with 2000× denser vacuum cores produce stronger echoes. Testable by Einstein Telescope and Cosmic Explorer.

**Prediction 7: Tangential peculiar velocities.** Predominantly tangential galaxy flow along void boundaries, with radial components suppressed relative to ΛCDM predictions. Testable by DESI, Euclid, and 4MOST.

**Prediction 8: EHT shadow fine structure.** Additional photon rings beyond the Kerr prediction, whose number and spacing encode the vacuum core geometry. SMBH formed at $z \sim 12$ should show different ring patterns from objects formed recently. Testable by next-generation EHT (space baseline).

**Prediction 9: CMB consistency.** Spatial curvature $\Omega_K > 0$ (small but positive, consistent with the Pole geometry). Low-ℓ power deficit from finite inflation duration. Testable by CMB-S4 and LiteBIRD.

**Prediction 10: Vacuum star electromagnetic counterpart.** Objects in the 2.2–4.0 $M_\odot$ range may produce electromagnetic emission from their chiral transition surface (no horizon to block it). A detected electromagnetic counterpart from a merger product above the OV limit would confirm the vacuum star prediction. Testable by multi-messenger campaigns.

**Prediction 11: Primordial gravitational-wave background with $r \approx 0.003$.** The Starobinsky $R^2$ recoil predicts a tensor-to-scalar ratio $r \approx 0.003$–$0.004$ and a tilt $n_t \approx -0.0004$. This is a single number, not a range. The nonlinear elastic regime near the Pole imprints characteristic departures from the power law on the longest scales. Detectable through B-mode CMB polarisation (LiteBIRD, CMB-S4), pulsar timing arrays (NANOGrav—the cosmological component distinguished by isotropy), LISA (millihertz, intermediate relaxation), and Einstein Telescope (upper limits at 1–100 Hz). A measurement of $r = 0.003 \pm 0.001$ would constitute direct detection of the Pole's structure.

**Prediction 12: Void dipole in $H_0$.** For each large void, the locally measured Hubble constant should exhibit a dipolar asymmetry: reduced on the near side (relaxation opposes Hubble flow), enhanced on the far side (relaxation adds to Hubble flow), neutral on the flanks. The amplitude scales linearly with void radius. Testable by stacking void-centric $H_0$ measurements from SDSS, DESI, and Euclid distance-ladder data. Absence of the dipole would refute localised relaxation.

**Prediction 13: Plateau foam topology and structural lexicon.** Junction angles of filaments at nodes should cluster around 120° and 109.47° (Plateau's laws). Filament thickness should scale as $(1+z)^2$. A finite set of repeating topological motifs (structural lexicon) should be identifiable across the full DESI volume. ΛCDM, built on Gaussian random initial conditions, predicts no preferred angles, no repeating motifs, and no characteristic cell size beyond the BAO scale. Detection of a structural lexicon would confirm the fixed-point attractor; its absence would support stochastic structure formation.

**What would falsify the programme**: (1) Detection of a dark matter particle. (2) Derivation of $\Lambda = 8\pi G\rho^{vac}$ from a fundamental principle. (3) $\alpha$ excluded in the range 0.001–0.01 at $>5\sigma$ by joint Planck+DESI+LSST analysis. (4) Tidal Love numbers measured to be exactly zero for objects above the TOV limit. (5) No correlation of $M^{dark}/M^{baryon}$ with distance from group centre for satellite galaxies. (6) Rotation curves measured smoothly beyond the predicted capture radius $r_c$ with no decline.

## 13. Ten Fundamental Problems Dissolved by a Single Separation

The separation of $\Lambda$ from $\rho^{vac}$—the single conceptual move at the foundation of this programme—dissolves eight problems that have haunted theoretical physics for decades. None required new particles, new forces, or new free parameters. All required only the recognition that an assumption made in 1967 was never a theorem.

### 13.1. The Vacuum Catastrophe (120 Orders of Magnitude)

**The problem.** Quantum field theory predicts a vacuum energy density $10^{120}$ times larger than the observed cosmological constant—the largest discrepancy in the history of physics.

**Dissolved.** The discrepancy presupposes the Zel'dovich identification $\Lambda = 8\pi G\rho^{vac}$. Once $\Lambda$ and $\rho^{vac}$ are recognised as physically distinct quantities—geometry versus field physics—the comparison that generates the $10^{120}$ factor is meaningless. The vacuum energy gravitates locally through $\alpha$, not globally through $\Lambda$. There is no catastrophe because there was never an identity.

### 13.2. Fine-Tuning and the Anthropic Principle

**The problem.** The cosmological parameters appear "tuned" for the existence of structure and life. Explanations invoke either a multiverse (unobservable) or an anthropic selection principle (unfalsifiable).

**Dissolved.** The cosmic web is the unique fixed point of the matter–vacuum–metric cycle (Theorem 10.3 of the companion Infodynamics paper). Its topology is determined by $\alpha$ and $\Lambda_0$ alone, independent of initial conditions. The parameters are not "tuned"; they are derived from the QCD sigma terms of the nucleon. Structure is not a lucky accident; it is a structural necessity of the self-consistent Block.

### 13.3. Dark Matter

**The problem.** Approximately 27% of the energy budget of the universe is attributed to a substance that has never been directly detected despite four decades of experimental search.

**Dissolved.** The "missing mass" is the gravitational effect of the quantum vacuum trapped inside bound systems. The isothermal vacuum profile reproduces flat rotation curves, the Tully–Fisher relation, the radial acceleration relation (with $a_0$ derived, not fitted), and the Bullet Cluster morphology. No new particle is required.

### 13.4. Dark Energy

**The problem.** Approximately 68% of the energy budget is attributed to a repulsive substance driving accelerated expansion. Its nature is unknown.

**Dissolved.** There is no repulsive substance. The observed acceleration is passive geometric relaxation of the spacetime membrane toward its flat ground state. The DESI evidence for dynamical dark energy ($w_a < 0$) is the deceleration of relaxation as the membrane approaches Attractor B. The "energy" was always geometry.

### 13.5. Singularities

**The problem.** General relativity predicts infinite density at the centres of black holes and at the origin of the universe. Physical laws break down.

**Dissolved.** The Kriger Limit replaces singularities with finite-density vacuum cores. The quantum vacuum, being a field that self-superposes rather than compresses, provides repulsive pressure at supranuclear densities ($w = -1$). The cosmological singularity is replaced by the Pole of Maximal Curvature—a smooth, finite-density topological extremum of the Block.

### 13.6. Heat Death

**The problem.** The second law of thermodynamics, extrapolated to the cosmos, predicts a future of maximal entropy, zero temperature gradients, and complete structural dissolution.

**Dissolved.** The end state of the universe is structural stillness, not thermal death. The megastructure is a gravitationally bound system whose binding energy exceeds the stress of relaxation. When the membrane reaches its ground state (Attractor B), the lattice of filaments and nodes remains permanently frozen. Effective complexity does not decrease; it converges to its maximum at the fixed point.

### 13.7. Quantum Gravity

**The problem.** General relativity and quantum mechanics are incompatible at the Planck scale. String theory, loop quantum gravity, and other approaches have produced no testable predictions in over forty years.

**Dissolved.** The incompatibility presupposes that quantum vacuum energy gravitates through the Einstein equations at full strength. In the *αLGQV* framework, QCD confinement sequesters vacuum energy inside the nucleon: the proton's internal vacuum does not create independent curvature (Section 7). GR governs the metric architecture; QCD governs the building material; the coupling α is the interface between them. The two theories are not in conflict; they operate on different aspects of the same Block Universe, connected by a single measurable constant derived from nuclear physics. No Planck-scale unification is required because the apparent conflict arose from forcing the full QFT vacuum energy into the gravitational source term—a step that was assumed, not derived.

### 13.8. Cosmic Inflation as an Ad Hoc Mechanism

**The problem.** The horizon problem, the flatness problem, and the monopole problem require a hypothetical epoch of exponential expansion driven by a hypothetical inflaton field with a hypothetical potential, none of which has been independently confirmed.

**Dissolved.** The Starobinsky potential is derived—not postulated—from the $R + R^2$ gravitational action as the elastic potential of spacetime (Paper #6). Inflation is the elastic recoil of maximally deformed spacetime, not a separate mechanism driven by a new field. The flatness of the universe is a consequence of topological closure and relaxation (Section 9), not of an inflaton. The horizon problem is dissolved by the coherence of the Block Universe at Attractor A, where all points were causally connected.

### 13.9. The Matter–Antimatter Asymmetry (Baryogenesis)

**The problem.** If equal amounts of matter and antimatter were produced in the Big Bang, some mechanism (CP violation, baryon number non-conservation, departure from thermal equilibrium—Sakharov's three conditions) must have preferentially destroyed the antimatter, leaving behind the matter we observe. Despite decades of experiment, no mechanism of sufficient magnitude has been identified. The baryon asymmetry remains unexplained within the Standard Model.

**Dissolved.** The problem presupposes a dynamical creation event: a moment at which matter and antimatter were "produced" from energy and subsequently separated by some unknown process. In the Block Universe, no such event occurs. The 4D manifold exists in its entirety, with its matter content as a structural property of the fixed point—not as a product of a process. Matter does not "arise from nothing." It is part of the self-consistent geometry of the Block, determined by the fixed-point conditions at Attractor A. The question "why is there more matter than antimatter?" assumes a symmetric initial state that was subsequently broken. In the Block, there is no initial state—there is only the structure. The asymmetry is not a deviation from a symmetric creation; it is a property of the manifold, as intrinsic as its topology.

Furthermore, annihilation itself acquires a functional role in this framework. In the atemporal structure of the Block, annihilation is not a "destruction" of matter but a constraint: it ensures that only self-consistent configurations—those that satisfy the fixed-point conditions of $\Phi_{cycle}$—persist as entries in the manifold. Configurations that would violate self-consistency (including exact matter–antimatter symmetry at all scales) are not "destroyed" in time; they simply do not appear in the fixed point. Annihilation is not a catastrophe; it is a selection principle—the geometric filter that determines which configurations belong to the Block and which do not.

### 13.10. The Expectation of Symmetry Itself

**The problem.** Many of the deepest puzzles in physics—baryogenesis, CP violation, the cosmological constant, the arrow of time—are formulated as "why is there asymmetry?" The implicit assumption is that the universe ought to be symmetric, and that any deviation requires explanation.

**Dissolved.** The expectation of symmetry is not a property of the universe. It is a property of the human mind (Kriger 2025). Symmetry preference is rooted in evolutionary pattern-hunger (survival advantage of detecting regularity), sexual selection (bilateral symmetry as a fitness signal), perceptual fluency (symmetric forms are processed with less cognitive effort), and cultural indoctrination (balance = good). The mind, shaped by these biases, projects its preference for regularity onto reality and then expresses surprise when reality fails to comply.

In physics, this projection has been elevated from a heuristic to a metaphysical principle (§10 of Kriger 2025: "From heuristic to metaphysical idol"). Noether's theorem connects continuous symmetries to conservation laws—but this does not mean that the universe is obligated to exhibit all possible symmetries. Conservation laws describe what persists under transformations that *happen to hold*; they do not prescribe that all conceivable transformations must be symmetries. The universe is under no obligation to be symmetric. It is under obligation only to be self-consistent—and self-consistency, as the fixed-point theorem demonstrates, does not require symmetry. It requires only that the matter–vacuum–metric cycle converge.

The dissolution of this meta-problem is perhaps the deepest of all. It does not merely solve a specific puzzle; it removes the cognitive lens that generates an entire class of puzzles. Once symmetry is recognised as a perceptual bias rather than a cosmic imperative, the questions "why is there asymmetry?" and "why is there something rather than nothing?" lose their metaphysical force. The Block Universe is what it is: a self-consistent manifold whose properties are determined by its own fixed-point conditions, not by the aesthetic expectations of its inhabitants.

In sum: ten problems, one conceptual move. The Zel'dovich identification of 1967 was never proved. Removing it, and replacing it with a coupling derived from QCD, dissolves eight of them directly. The ninth (baryogenesis) is dissolved by the Block Universe's elimination of the "creation event" that generates the problem. The tenth (the expectation of symmetry) is dissolved by recognising that the deepest assumption—that the universe should be symmetric—is a cognitive artefact, not a physical law. Occam's razor is satisfied not by postulating fewer entities but by *demonstrating* that fewer entities suffice, and by *auditing* the cognitive biases that made additional entities seem necessary.

## 14. The Relativity of Cosmic Age and the Primacy of Structural Connectivity

### 14.1. The Age of the Universe Has No Single Value

The statement "the universe is 13.8 billion years old" is a coordinate artefact. In the Block Universe, different regions of the 4D manifold experience radically different proper times relative to a comoving observer:

**Inside supermassive black holes** formed at $z \approx 12$, proper time is nearly frozen. The internal state of these objects has aged by at most a few years since the epoch of collapse. For the vacuum core, it is still the beginning. The "age" of the universe inside a primordial black hole is measured in years, not billions of years.

**Inside deep voids**, the situation is reversed. The gravitational potential in the centre of a large void is a local minimum: all mass (filaments, clusters) lies at the boundaries. Despite the colossal total mass of the vacuum field integrated over the void volume, the vacuum does not self-collapse (Section 4.2)—it is a field that self-superposes, not a gas that contracts. Therefore the gravitational time dilation at the void centre is *less* than in the matter-dominated environment where we reside. Clocks in the void centre tick faster. A rough estimate: for a void of radius $R \approx 50\ h^{-1}$ Mpc with boundary overdensity $\delta \approx 1$, the integrated Shapiro delay gives a fractional time gain of order $\Delta t/t \sim \Phi/c^2 \sim 10^{-5}$. Over 13.8 Gyr, this accumulates to $\sim 10^5$ years. The effect is modest on a cosmic scale, but it is real and directional: void centres are always older than their boundaries. In deeper gravitational analyses incorporating the full void hierarchy (voids within voids), the cumulative effect grows.

The "age of the universe" is therefore a position-dependent quantity in the Block. The 13.8 Gyr figure is our local proper time at a specific location in the filament network. It is not a property of the universe; it is a property of our worldline.

### 14.2. The 90-Billion-Light-Year Megastructure

The observable universe has a comoving diameter of approximately 93 billion light-years—far larger than the 13.8 billion light-years that light has had time to travel. In ΛCDM, this discrepancy is attributed to the expansion of space during the photon's journey. In the relaxation framework, the explanation is identical in its mathematics but different in its ontology: the metric has relaxed (smoothed out) during the photon's propagation, so the comoving distance between emission and reception has grown.

We will never see the edges of the megastructure. The particle horizon is finite; the structure may not be. But—and this is the central point—**the megastructure will not fly apart**. The fact that we cannot observe its entirety with light does not mean it lacks coherence. It is a gravitationally bound system. Bound systems do not require light to maintain their integrity. The Earth does not fly apart because we can see it; it holds together because of gravity. The cosmic web holds together for the same reason, scaled up by a factor of $10^{26}$.

### 14.3. Structural Connectivity vs. Causal Connectivity: The Domino Principle

Modern cosmology has inherited a bias from special relativity: the assumption that the only meaningful connectivity is *causal* connectivity—the ability to exchange light signals within a finite time. Regions outside each other's light cones are considered "disconnected." This is correct for signal transmission but profoundly misleading for structure.

Consider a row of dominoes. No domino "sends a signal" to the last domino in the row. Yet the structure ensures that if the first falls, the last will fall. The connectivity is **structural**, not causal. Each domino is connected only to its immediate neighbour; the chain transmits the perturbation without any single element "knowing" about the endpoint.

The cosmic web operates on precisely this principle. Filaments connect neighbouring clusters. Clusters anchor neighbouring filaments. The lattice is a continuous chain of curvature wells, each influencing its immediate neighbours through the metric. A perturbation at one node propagates through the lattice not at the speed of light but at the speed at which the metric adjusts—which, for the fixed-point topology, is irrelevant, because the fixed point is already determined by the global structure of the attractor, not by the sequential propagation of signals.

This is the deepest consequence of the Block Universe: the lattice does not *become* coherent by exchanging signals across its extent. It *is* coherent because it is a single static structure in 4D, determined by the fixed-point conditions at the Pole. The domino principle is the spatial analogue of what the temporal anchors (black holes) do in time: they ensure coherence not by transmitting information but by being structurally connected elements of a single self-consistent manifold.

A second, even deeper channel of non-causal connectivity exists: **quantum entanglement**. Particles created together in the dense primordial epoch near Attractor A—photons, baryons, leptons produced in the same interaction vertices—were subsequently carried apart by the relaxation of the metric. Over 13.8 billion years of comoving separation, many entangled partners have been swept beyond each other's particle horizons. They are causally disconnected: no light signal can ever pass between them again. Yet their quantum correlations persist. Entanglement is not destroyed by spatial separation; it is destroyed only by decoherence with a local environment.

In the Block Universe, this is not paradoxical. The entangled state is a single entry in the 4D manifold—a correlation written into the structure at the Pole, carried forward along the temporal gradient, and preserved in the static geometry of the Block regardless of whether the partners remain within each other's light cones. The Block does not require signals to maintain correlations; the correlations are part of the manifold's structure, exactly as the lattice of filaments is part of the manifold's curvature.

The universe is therefore quintuply connected beyond causal horizons: (i) **gravitationally**, through the domino chain of curvature wells in the filament lattice; (ii) **temporally**, through the frozen vacuum cores of supermassive black holes anchored at the Pole; (iii) **quantum-mechanically**, through the entanglement of particles born together and separated by relaxation; (iv) **photonically**, through the CMB and all radiation fields, whose photons have zero proper time; and (v) **vibrationally**, through the primordial gravitational-wave background—ripples on the membrane itself, emitted at the Pole and propagating with zero proper time to our epoch, the deepest structural threads in the Block.

### 14.4. The Fourth Channel: Photons as Structural Threads

Special relativity states: at $v = c$, proper time $d\tau = 0$. This is exact, not approximate. For any photon, between the moment of emission and the moment of absorption, zero proper time elapses. The photon does not "travel" from past to present. For the photon, emission and absorption are the same event. What we perceive as 13.8 billion years of flight is a coordinate artefact of our timelike worldline. In the photon's own frame, there is no duration.

A CMB photon was emitted at $z \approx 1100$, roughly 380,000 years after the Pole. By our clocks, 13.8 billion years have elapsed. By the photon's clock: zero. The photon is a one-dimensional object in 4D, stretched from the surface of last scattering to our detector, with zero proper length along the time axis. It is not a messenger from the past. It is a **structural thread**—a seam in the fabric of the Block, connecting the early and late epochs with zero temporal gap.

The CMB is not a relic. It is not an echo. It is 400 photons per cubic centimetre, each a thread with zero proper time, collectively forming a fabric that stitches the Block together from the epoch of recombination to the present. Every cubic metre of the universe contains hundreds of billions of these threads. The early universe is not "out there, far away." For the photons that connect us to it, there is no distance. The surface of last scattering is here—because for the photon, it never left.

And not only the CMB. Every photon currently in flight—from a quasar at $z = 7$, from the Sun eight minutes ago, from a lamp on the desk nanoseconds ago—is a structural thread with zero proper time, connecting its emission event to its absorption event with no temporal gap. The universe is *sewn together by light*. Every photon is a stitch. The Block is not merely connected by gravity, by frozen cores, and by entanglement. It is woven with light, in every direction, at every scale, with every thread having zero proper duration.

This dissolves the horizon problem without inflation. The standard question—"how did regions separated by more than two degrees on the CMB sky reach the same temperature, if they could not exchange signals in 380,000 years?"—presupposes that thermal equilibrium requires causal contact. In the Block, equilibrium is not a process that requires signal exchange. It is a property of the fixed point. The geometry at the Pole is the unique self-consistent solution of the matter–vacuum–metric cycle. Homogeneity is built into the solution, not achieved by a mechanism. The CMB photons we observe are not evidence that equilibrium *was reached*. They are the equilibrium itself, delivered to our detectors through threads of zero proper time.

Entangled particles separated by metric relaxation present a related insight. A pair born together at the Pole, one subsequently captured by an SMBH at $z \approx 12$ (proper time nearly frozen) and the other drifting into a void (proper time running slightly faster than ours), remain entangled despite their radically different proper-time histories. In the Block, this is not paradoxical: entanglement is not a correlation maintained through time. It is a topological property of the manifold—a single entry in the 4D structure, written at the Pole and invariant along both worldlines regardless of how much proper time has elapsed along each. The EPR "paradox" dissolves: no "spooky action at a distance" is needed because no action occurs. Correlation is a property of structure, not of communication. Two ends of a rod do not "communicate" their rigidity. They are one rod. Entangled particles across the universe are one object in 4D, projected onto our 3D perception as two.

### 14.5. The Fifth Channel: The Gravitational Relic Background

The CMB connects us to $z \approx 1100$—the epoch when the universe became transparent to photons. Before that, the plasma was opaque. No photonic threads penetrate earlier. But spacetime is transparent to gravitational waves *always*. Gravitational waves are ripples on the membrane itself. The membrane cannot be "opaque" to its own oscillations. Therefore, the primordial gravitational-wave background—generated at the Pole, at the moment of maximum deformation—constitutes the deepest structural threads in the Block, connecting us directly to Attractor A.

In the αLGQV framework, the spectrum of this background is computable with zero free parameters. The Starobinsky $R^2$ inflation (the elastic recoil of the maximally deformed membrane) predicts a tensor-to-scalar ratio $r \approx 0.003$–$0.004$. This is a single number, not a range: it follows from the scalaron mass M, which is fixed by the CMB amplitude $A_s$. The spectral tilt is $n_t = -r/8 \approx -0.0004$—nearly scale-invariant, with a specific predicted deviation. The nonlinear elastic corrections near the Pole (where $m^2_{eff}(R) = M^2R/R_{Pole}$ deviates from the plateau approximation) imprint a characteristic departure from the pure power-law spectrum on the longest scales.

Competing frameworks cannot match this specificity. String theory cannot compute the spectrum because the vacuum is not determined. Loop quantum gravity predicts a bounce spectrum that depends on the adjustable Immirzi parameter. Standard ΛCDM admits r from 0 to 0.3 depending on the choice of inflaton potential. The αLGQV prediction is one number.

**Detection methods.** (1) **B-mode polarisation of the CMB**: gravitational waves passing through the primordial plasma at $z \approx 1100$ imprint a curl pattern in CMB polarisation. LiteBIRD (launch late 2020s) and CMB-S4 are designed to detect $r \approx 0.003$ at $>3\sigma$. (2) **Pulsar timing arrays**: NANOGrav, EPTA, and PPTA have detected a stochastic background in the nanohertz range. In the αLGQV framework, part of this signal is cosmological (from membrane relaxation), distinguishable from astrophysical sources by its isotropy. (3) **LISA**: sensitive to the millihertz range, probing intermediate stages of relaxation with a spectrum determined by the running stiffness. (4) **Einstein Telescope and Cosmic Explorer**: the 1–100 Hz range, where the cosmological background is extremely faint but whose predicted level is parameter-free.

If LiteBIRD or CMB-S4 measures $r = 0.003 \pm 0.001$, this will constitute a direct detection of the Pole's structure—the vibrational fingerprint of maximal membrane deformation, carried to our detectors through threads of zero proper time. The gravitational relic background is the fifth and deepest seam of the Block: photons connect us to recombination; entanglement to the birth of particles; black holes to $z \approx 12$; filaments to neighbouring nodes; and the gravitational background to the Pole itself.

### 14.6. Against the Tyranny of Light

We assign excessive ontological weight to electromagnetic radiation. Because black holes do not emit light, we call them "holes"—absences, voids in being. But a black hole has internal structure: a vacuum core, a chiral transition shell, a magnetic architecture (Paper #22 of Volume II). The fact that we cannot see this structure does not diminish its physical reality.

Similarly, because we cannot see the far side of the megastructure, we treat it as epistemically inaccessible—and from this, slide into treating it as physically uncertain. But the megastructure is a bound system. Its far side is connected to our side by the same lattice of filaments and anchors. We do not need to see it to know it is there, any more than we need to see the far side of the Earth to know it exists.

The irony is precise: light is simultaneously less important than standard cosmology assumes (it is not the only or deepest channel of connectivity) and more important (photons are structural threads that sew the Block together with zero proper time). We have treated light as a messenger—a carrier of information about distant places. It is that. But it is also a *structural element*: a zero-proper-time seam in the 4D manifold, as fundamental to the architecture of the Block as a filament or a temporal anchor.

The *αLGQV* framework replaces the tyranny of the light cone with the democracy of the metric. What matters is not whether a photon can reach us from a given region, but whether that region is part of the same self-consistent curvature network. If it is—and the contraction mapping theorem guarantees that the entire lattice is a single fixed point—then it is as real and as connected as the chair we sit on. The universe is not an island of visibility in an ocean of unknowability. It is a structure, held together by gravity, entanglement, geometry, and light.

### 14.7. Hypercoherence

Standard coherence is a single-channel property. A laser is coherent: photons are in phase. A superconductor is coherent: electron pairs share one quantum state. Disrupt the mechanism—heat the superconductor, defocus the laser—and coherence vanishes.

The Block Universe is not coherent. It is **hypercoherent**: five independent channels of structural connectivity, each operating on different physics, each non-causal, each an irreducible consequence of the same fixed point. Gravitational (metric curvature). Temporal (frozen vacuum cores). Quantum (entanglement as topological invariant). Photonic (zero-proper-time threads). Vibrational (primordial gravitational-wave background). Five mechanisms. Five scales. All simultaneously, redundantly, inseparably.

This is not engineering redundancy (three copies of one system). It is structural necessity. Each channel is an *inevitable consequence* of the fixed point. Gravity exists because the metric exists. Entanglement exists because quantum fields exist. Photonic threads exist because massless excitations propagate at the membrane speed. Temporal anchors exist because collapse halts at the Kriger density. The gravitational background exists because the Pole had finite deformation. Remove any one—and you must remove the fixed point itself. The Block cannot exist with four channels out of five, because the fifth follows from the same equations as the other four.

**Definition.** Hypercoherence is the property of a system in which multiple independent channels of connectivity are not supplementary but *irreducible consequences of a single fixed point*, so that the coherence of the system cannot be disrupted by eliminating any subset of channels, because each channel exists with the necessity that follows from the existence of the others.

Ordinary coherence can be destroyed. Heat a superconductor—coherence vanishes. Hypercoherence cannot be destroyed without destroying the system. To remove one channel, one must alter the fixed point. But the fixed point is unique. Alter it—and the Block ceases to exist. Not weakened. Not degraded. Erased.

This dissolves the decoherence problem. Quantum decoherence—the loss of quantum correlations through interaction with the environment—destroys one channel (quantum). But the Block remains connected through the other four. Decoherence is not a loss of coherence. It is a *transfer of dominance*: from quantum coherence at microscopic scales to gravitational, photonic, and vibrational coherence at macroscopic scales. The classical world does not "lose" quantum correlations. It expresses the same connectivity through channels that dominate at its scale.

Hypercoherence also answers Wigner's question—"why is mathematics so unreasonably effective in the natural sciences?" The brain is a structure persisting on a biological substrate. The universe is a structure persisting on a physical substrate. Both obey the same persistence laws. Mathematics is effective not because reality "obeys" mathematics, but because both mathematics and reality are projections of the same fixed point. A brain modelling the universe is a part of the Block building an internal representation of the Block. The model works because the model and the modelled are one structure, observing itself from within.

If hypercoherence is a property of the unique fixed point, then it is not optional. Any persisting universe is hypercoherent. A universe without multiple irreducible channels of connectivity does not persist—it does not pass through the Banach filter—it does not close. Hypercoherence is not a feature of our universe. It is the condition of existence of *any* universe.

### 14.8. Layered Uncertainty and Dual Redundancy

The hypercoherent Block exhibits a second structural principle that operates orthogonally to the five channels: a strict alternation of indeterminacy and emergent determinacy across scales. At the subatomic scale, the quantum vacuum is maximally uncertain—virtual fluctuations, Kolmogorov noise, no predictable structure. From this uncertainty, the nucleon emerges as a fixed point of absolute determinacy: invariant mass, invariant charge, invariant spin. At the interstellar scale, gas and dust are again uncertain—turbulent, statistical, chaotic. From this uncertainty, gravitational collapse produces stars and planets: determinate, structured, functionally organised. At the void scale, the free vacuum is featureless—pure background, Kolmogorov noise. From this uncertainty, the megastructure emerges as the unique fixed point of the matter–vacuum–metric cycle: determinate, topologically specified, algorithmically compressible.

This alternation is not accidental. It is structurally necessary. The uncertainty layers provide **adaptivity**: degrees of freedom that allow the system to accommodate perturbations, absorb shocks, and explore configuration space. The determinacy layers provide **structural coherence**: persistence, memory, and functional load-bearing. A system that was all determinacy would be a crystal—rigid, brittle, incapable of responding to perturbation. A system that was all uncertainty would be a gas—formless, structureless, incapable of supporting function. Persistence requires both: flexible joints between rigid bones.

The principle extends to the redundancy structure. Each physical parameter serves *multiple functions* (minimum entities, maximum functions, Section 5.4). The coupling α simultaneously determines rotation curves, the $S_8$ tension, the cosmological constant, and the seed formation window. The scalaron mass M simultaneously determines inflation, the Kriger density, the mass hierarchy, and the gravitational-wave background. Conversely, each structural feature is secured by *multiple parameters*: the cosmic web is maintained by α, $\Lambda_0$, M, and $f_b$ acting in concert. This is dual redundancy: every element has many roles, and every role has many elements. Neither can be removed without cascading failure across the entire fixed point.

The alternation of uncertainty and determinacy, combined with dual redundancy, constitutes the full architecture of persistence. The uncertainty layers are the system's *immune response*—its capacity to absorb novelty without structural collapse. The determinacy layers are its *skeleton*—

its capacity to maintain form across the entropic gradient. Together, they make the Block not merely stable but **antifragile**: a system that becomes more coherent under perturbation, because every perturbation drives the contraction mapping closer to its fixed point ($k \approx 0.004$, Section 5.4).

## 15. Relation to the Unified Structural Theory of Complex Systems

The cosmological architecture presented here constitutes one major application of the broader *Unified Structural Theory of Complex Systems* (Kriger 2026b), which demonstrates that the formal constraints governing persistence—the capacity of a system to maintain structural viability over time—are substrate-independent. The same mathematical structures (fixed-point theorems, viability sets, cyclical closure, attractor dynamics) that govern the cosmic web also govern cognitive architecture, institutional dynamics, and biological self-organisation.

The Block Universe with its two attractors is a specific instance of the general principle: any self-consistent persistent structure must be bounded by attractor states that define its domain of viability. The Kriger Limit is a specific instance of the Viability Mismatch Law: collapse to singularity would violate the persistence conditions of the vacuum field. The coherence of the megastructure is a specific instance of the four laws of self-organising systems (Cooperation, Viability, Interference, Observability). The mathematical apparatus is identical: the cosmic lattice and a neural network are governed by the same fixed-point topology; the Kriger Limit and the breakdown of a cognitive model under extreme stress are governed by the same viability boundary.

## 16. Programme of Proof: Papers Required Beyond Volume I

The architecture presented in this paper rests on two categories of results: those already established in Volume I (Papers #1–#17, plus the companion Infodynamics paper), and those requiring further derivation. This section maps every major claim to the paper(s) that prove it, distinguishing between what is done and what remains to be done. Papers marked with † are planned for Volume II; papers marked with ‡ are proposed new investigations.

### 16.1. Claims Established by Volume I (Papers #1–#17)

The following claims require no further proof; they are derived in the published monograph:

$\Lambda \neq \rho^{vac}$ (five independent formalisms) — Papers #1, #1a. The vacuum–matter ansatz $\rho^{vac}(\rho^{m}) = \Lambda_0 - \alpha\rho^{m}$ — Papers #2, #3. The QCD derivation of $\alpha = 0.005$ from sigma terms — Paper #3a. Nonlinear self-screening (factor ~3) from N-body simulations — Papers #8, #9. Starobinsky potential derived from $R + R^2$ action — Paper #6. Flat rotation curves, Tully–Fisher, RAR, Bullet Cluster from vacuum capture — Paper #12. Cosmic web as unique fixed point of $\Phi_{cycle}$ — Paper #13. Type Ia supernovae unaffected ($\Delta M_{Ch}/M_{Ch} \sim 10^{-28}$) — Paper #16. Block Universe and Copernican principle — Paper #17.

### 16.2. Claims Established by the Companion Infodynamics Paper

Monotonic growth of effective complexity $K^{eff}$ toward the fixed point (conditional on Condition C). Simulability $S = 1 - k \approx 0.996$. Two information bits (infodynamic and QCD). Topological universality of cost-minimising networks. Eight falsifiable predictions distinguishing the model from ΛCDM. All derived in Kriger (2026d).

### 16.3. Claims Requiring Volume II Papers

**The Kriger Limit and vacuum incompressibility** (Section 4). The hierarchy Eddington → Chandrasekhar → Oppenheimer–Volkoff → Kriger requires derivation of the vacuum pressure equation of state at supranuclear densities. — *Paper #20†* (The Third Step: Vacuum Pressure as the Stabiliser of Gravitational Collapse).

**Internal structure of collapsed rotating objects** (Section 4.4). The toroidal vacuum core, chiral transition shell, layered structure. — *Paper #21†* (The Internal Structure of Collapsed Rotating Objects).

**Magnetic architecture of the vacuum core**. Flux conservation, type-II superconductivity, CFL Meissner effect, tokamak stability. — *Paper #22†* (Magnetic Architecture of the Vacuum Core).

**Two timescales inside collapsed objects** (Section 14.1). Proper time $d\tau/dt_\infty \approx 0$ but $> 0$; relict vacuum frozen at $z \approx 12$. — *Paper #23†* (Time Inside: The Two Clocks of a Collapsed Object).

**Event horizon as observer property, not object property**. No-hair theorem violated by vacuum cores. — *Paper #24†* (The Event Horizon Reconsidered).

**Gravitational-wave predictions** (Section 12). QNM overtones, echoes ($\Delta t \sim$ ms), finite tidal Love numbers. — *Paper #25†* (Gravitational Waves from the Interior: Three Channels of Information).

**Supermassive seeds at $z > 10$ and the 2000× vacuum enhancement** (Sections 4.3, 14.1). Vacuum-assisted direct collapse, characteristic masses $10^{4–5}$ M⊙. — *Paper #26†* (Vacuum-Assisted Direct Collapse and the Formation of Supermassive Seeds).

**The mass desert** (300–$10^4$ M⊙). Two non-overlapping formation channels; gravitational recoil prevents sequential mergers from bridging the gap. — *Paper #27†* (The Mass Desert: Why Intermediate-Mass Black Holes Are Rare).

**Impossibility of wormholes**. Eight independent arguments from the gravitating vacuum. — *Paper #28†* (Against Wormholes).

**The Pole of spacetime** (Section 2.1). Finite density $\sim 10^{73}$ kg/m³, no "before," no bounce. — *Paper #29†* (The Pole of Spacetime: No Singularity, No "Before," No Bounce).

**The mass hierarchy problem** ($10^{55}$ between inflationary and late-time scales). Candidate solution: $m^2_{eff}(R) = M^2 \times R/R_{Pole}$ (Section 8.8). — *Paper #31†* (The Mass Hierarchy Problem). Status: *candidate solution; requires rigorous derivation from running $R^2$ coefficient*.

**Complete falsification programme for Volume II**. Concrete numbers, thresholds, and timelines for LIGO, LISA, Einstein Telescope, EHT, JWST, CMB-S4, eROSITA. — *Paper #32†* (Observational Signatures).

### 16.4. Claims Requiring New Investigations

**Relaxation vs. expansion: rebound effect and DESI data** (Section 3.3). Derivation of $w_{eff}(z)$ from the elastic potential of the metric; comparison with DESI $w_0$–$w_a$ contours. — *Paper #33‡* (Metric Relaxation and the DESI Evidence for Dynamical Dark Energy).

**Peculiar velocity field of filament galaxies** (Section 5.3). N-body extraction of tangential vs. radial velocity components relative to void surfaces; comparison with ΛCDM predictions. — *Paper #34‡* (Peculiar Velocities in the Gravitating Vacuum: Tangential Flow Along Void Boundaries).

**Algorithmic compressibility of the DESI cosmic web** (Section 10.4). Computation of $K^{eff}$ at successive redshift shells; morphological census (V, S, F, N, T classification); comparison of effective complexity with biological and AI networks. — *Paper #35‡* (The InfoUniverse: Effective Complexity of the Cosmic Web from DESI Data).

**Baryogenesis as a fixed-point property** (Section 13.9). Formal derivation of matter–antimatter asymmetry from the self-consistency conditions of $\Phi_{cycle}$; annihilation as a geometric filter. — *Paper #36‡* (Matter–Antimatter Asymmetry as a Structural Property of the Block Universe).

**Quantum entanglement as structural connectivity** (Section 14.3). Formal treatment of trans-horizon entanglement in the Block; comparison with ER=EPR programme; entanglement as a topological invariant of the 4D manifold. — *Paper #37‡* (Entanglement Beyond the Horizon: The Third Channel of Structural Connectivity).

**Symmetry as cognitive artefact** (Section 13.10). Formal extension of the cognitive audit (Kriger 2025) to Noether's theorem, gauge invariance, and CP violation; demonstration that the "symmetry puzzles" of physics arise from projecting a perceptual bias onto the cosmos. — *Paper #38‡* (The Cognitive Origin of Physical Symmetry Principles: From Heuristic to Metaphysical Idol).

**Lattice QCD gravitational form factors** (Section 7). Computation of the gravitational form factors A(t), B(t), D(t) of the nucleon with separate treatment of the scalar condensate component, to confirm or refute the $w = -1$ assignment for the chiral condensate shift (Proposition 5.1 of the

Infodynamics paper). — *Paper #39‡* (Gravitational Form Factors of the Nucleon and the Equation of State of the Chiral Condensate). Status: requires lattice QCD collaboration.

**Fermionic screening and the gravity interface** (Section 7.4). Quantitative model of how Pauli exclusion suppresses vacuum modes inside atoms; computation of the screened vacuum fraction as a function of atomic number; derivation of α as the surviving fraction after the screening cascade. — *Paper #40‡* (Fermionic Screening of Vacuum Gravity: The Interface Between Quantum Mechanics and General Relativity).

**Derivation of Λ from QCD parameters** (Section 8.7). Rigorous derivation of $\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6/m_{Pl}^4$ from the self-consistency conditions of the Block; demonstration that Λ is the unique eigenvalue of the 4D manifold given M, α, and $f_b$; numerical verification against Planck data. — *Paper #41‡* (The Cosmological Constant as a QCD Observable: A Parameter-Free Derivation of Λ from Nuclear Physics).

**Nonlinear ansatz at extreme densities**. Extension of $\rho^{vac}(\rho^m)$ beyond the linear regime using $a_2$ Schwinger–DeWitt coefficients; chiral phase transition at $\rho \sim 3$–$5\ \rho_{nuclear}$. — *Paper #19†* (The Nonlinear Vacuum: Beyond the Linear Ansatz at Extreme Densities).

In total: the architecture of this paper is supported by 19 established papers (Volume I), one companion paper (Infodynamics), 15 planned papers (Volume II, Papers #18–#32), and 9 proposed new investigations (Papers #33–#41). The mass hierarchy problem (Paper #31) has a candidate solution (Section 8.8) pending rigorous derivation. The cosmological constant is derived from QCD parameters with a ratio of 1.65 to the observed value (Section 8.7, Paper #41). All other claims are either proved, derived with stated assumptions, or mapped to specific future papers with defined scope and method.

## 17. Conclusion

We have presented a unified architecture of the Block Universe as a 4-dimensional static manifold bounded by two topological attractors. Cosmic expansion is reinterpreted as passive geometric relaxation. Singularities are eliminated by the Kriger Limit—the intrinsic incompressibility of the quantum vacuum field. Supermassive black holes function as temporal anchors; filaments as spatial connectors. The entire megastructure is a single bound system—a dynamic lattice sliding along the relaxing membrane.

The vacuum's gravitational influence obeys a strict scale hierarchy: negligible at stellar scales, dominant at galactic scales, architectural at cosmological scales. The internal vacuum structure of the nucleon—including the strange-quark sea essential for nuclear fusion—is an invariant of the Block, unchanged from the primordial epoch. The same sigma terms that produce galactic rotation curves make stars shine.

The model uses no new particles, no new forces, and no free parameters beyond those derived from QCD. It completes the programme that Einstein's general relativity began: a purely geometric cosmology in which all dynamics arise from the elastic properties of the spacetime metric and the gravitational weight of the quantum vacuum.

The century-old conflict between general relativity and quantum physics is dissolved by recognising that QCD confinement sequesters vacuum energy inside the nucleon: the proton's internal vacuum does not create independent curvature. GR governs the architecture; QCD governs the building material; α is the interface. The experienced arrow of time arises from the entropic bias of the Ledger—the exponential cost of un-committing entangled states in the atemporal DAG of quantum configurations—reconciling the static Block with the dynamic experience of temporal flow.

The effective complexity of this architecture—measured through the algorithmic compressibility of its lattice—places the cosmic web in the same informational regime as biological networks and trained AI systems. The universe is an InfoUniverse: its megastructure is not a thermodynamic accident but a functional, self-preserving information architecture governed by three principles—falsifiability, simulability, and functionality—that any adequate scientific description must satisfy simultaneously.

The universe may be simpler than we have allowed ourselves to imagine: a self-consistent elastic manifold, deformed once at its pole, relaxing ever since, and held together by the incompressible memory of its own quantum vacuum.

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